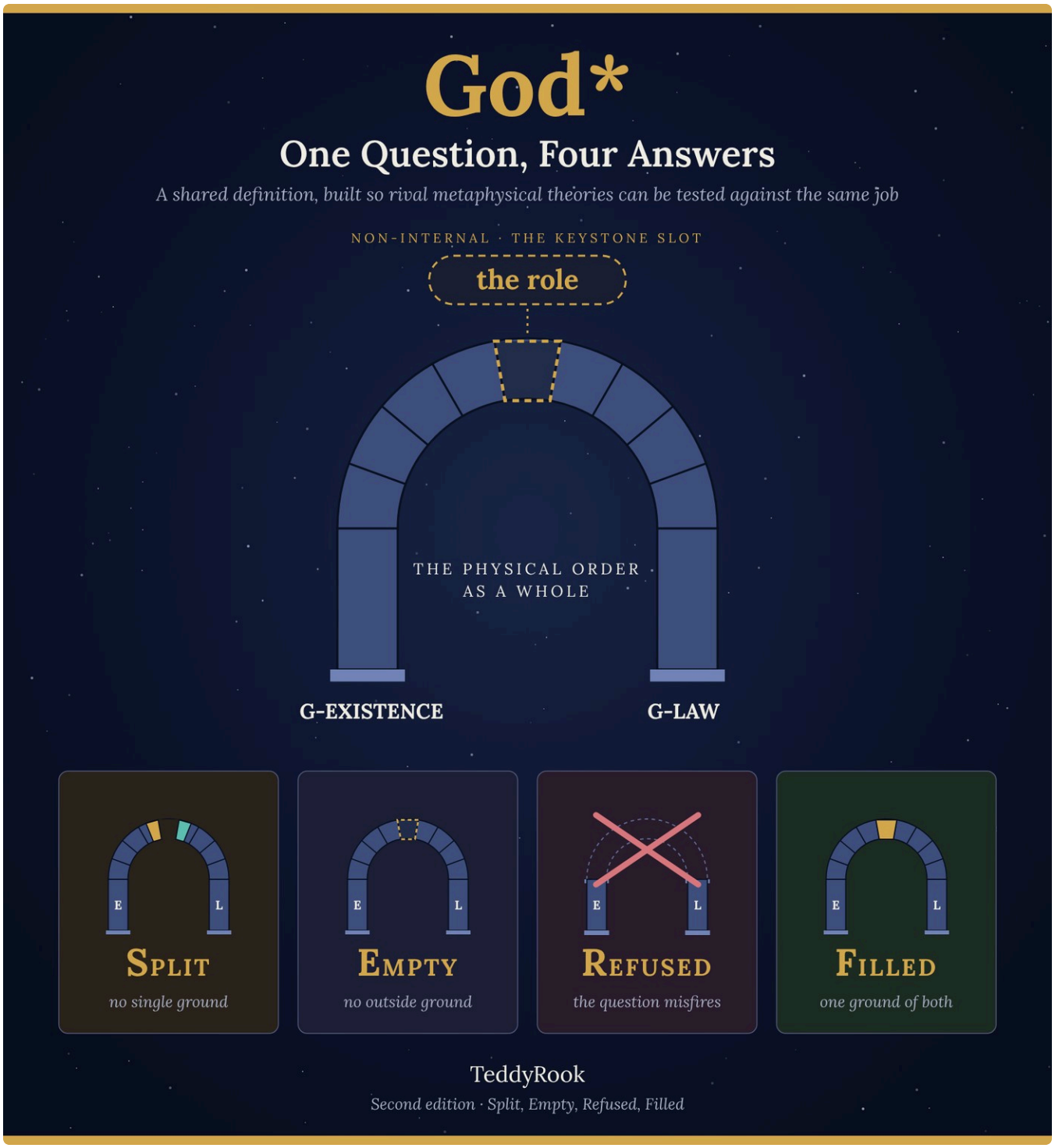
God*: One Question, Four Answers

Nov 10, 2026 · @TeddyRook



The Ground of the Physical Order as a Shared Vocabulary: A First-Principles Construction of a Role, Its Occupants, and Four Verdicts for Comparing Metaphysical Theories across Creeds

Second edition, rewritten from first principles

A monograph in the form of a doctoral dissertation

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Field: Metaphysics · Philosophy of Religion · Comparative Philosophy

Status: Working draft for self-study, and the basis for a future journal article or doctoral proposal. It has not been submitted to or examined by any university. References were compiled from memory and must be verified before formal use.

> "The first question which we have a right to ask will be: Why is there something rather than nothing?" — G. W. Leibniz, *Principles of Nature and Grace* §7 (1714)

> "The eternal mystery of the world is its comprehensibility." — Albert Einstein, "Physics and Reality" (1936)

> "Then was neither non-being nor being … He who surveys it in the highest heaven, he surely knows — or perhaps he does not know." — *Rig Veda* 10.129

Leibniz asks why there is a world. Einstein asks why it is orderly. The hymn admits the answer may be out of reach. This book builds one precise question that contains all three.

Abstract

People who argue about God often argue past each other, because the word "God" names different things in different mouths. This book builds, one step at a time, a single question that believers, atheists and members of non-theistic traditions can all be scored against. The question uses a starred technical term, **God***. God* is not the name of a being. It is the title of a role: *the terminal non-internal ground* of the physical order as a whole*, where that whole has two sides. **G-existence** is the fact that there is a physical order at all. **G-law** is the fact that it is lawful.

Part I constructs the question from nothing assumed. Each chapter teaches one idea from scratch and adds one piece to the definition: the two facts to be explained, the kinds of "because," the ways explanations can end, the line between inside and outside the physical order, the meaning of "the whole," and the logic of titles and roles.

Part II derives the possible answers. There are exactly four, and their initials spell **SERF**. **Split:** something outside grounds at least one side, but no single thing grounds both. **Empty:** nothing outside grounds either side. **Refused:** the view will assert neither that something grounds the order nor that nothing does. **Filled:** one thing outside the order grounds both sides. Three short tests decide between them.

Part III reads twenty-six centuries of Greek, Abrahamic, Indian and Chinese thought as evidence that these distinctions are real. Part IV tests the framework on hard cases, on four deep case studies (Aquinas, Nāgārjuna, Zhu Xi, Oppy), and on a real debate, and proposes a study of whether different scorers reach the same verdicts. Part V places the framework in the scholarly literature, answers objections, and states its limits.

The claim of originality is narrow. The ingredients are old. What is offered is an instrument: a two-sided role, a closed set of four answers, and explicit tests for applying them.

How this edition is built

The first edition taught the background, then stated the framework, then repaired it. This edition never needs the repairs, because each piece is built correctly the first time. Every chapter in Part I ends with a line headed **Built so far**, which shows the definition growing. By the end of Chapter 7 it is complete.

Four things are new, and each fixes a weakness found on review of the first edition.

1. **A sharper test for "internal."** An item is internal if citing it only *relocates* the question (Chapter 5). This handles simulated universes properly.
2. **An assent test for Refused.** A view is Refused only if it will affirm neither "something grounds the order" nor "nothing does" (Chapter 9).
3. **A defence of exactly two sides** (Chapter 2).
4. **A split between core and extensions** (Chapter 11), so the framework stays simple at its centre.

Appendix E lists every change from the first edition.

Reader's guide

The book assumes no background. Three devices help.

- **Ladder boxes.** Where an idea is known to be hard, a boxed “Ladder” explains it four times. **Rung 1** is for a curious ten-year-old. **Rung 2** is for a high-school student. **Rung 3** is for an undergraduate philosophy major. **Rung 4** is for a graduate student or researcher.
- **Primer pointers.** A name followed by → A.n has a short primer in Appendix A.
- **Key-idea and Built-so-far lines.** These state what to carry forward.

Notation.

- “God*” always means the role, never a particular god.
- **P*** is the physical order as a whole.
- **E** is G-existence: the fact that there is such an order.
- **L** is G-law: the fact that it is lawful.
- **Ground*** is the dependence relation defined in Chapter 3.
- Verdicts are written S, E, R, F, for Split, Empty, Refused, Filled. The order spells SERF. Context always makes clear whether E means the Empty verdict or the existence fact.

Shortcuts.

- For the framework only, read the Built-so-far lines, then Chapters 7, 8 and 22.
- For the lesson, read in order.
- For the question of publication, read Chapter 19 and Appendix D.

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Part I — Building the question

Part I assumes nothing. It starts from the most ordinary observations and adds one idea per chapter until a precise question stands complete. None of the individual ideas is original. The construction is.

Chapter 1 — Metaphysics and the ladder of "why"

1.1 What metaphysics is

"Metaphysics" began as a librarian's label. Aristotle wrote a set of treatises on the most general features of reality (→ A.4). About three centuries later an editor, traditionally Andronicus of Rhodes, placed them *after* the *Physics*: in Greek, *ta meta ta phusika*. The label stuck and became the name of a subject.

Today metaphysics is the study of what there is and what it is like at the most general level. It asks what it is to exist, what makes a thing the same thing over time, what causation is, and what makes something necessary or merely possible.

Key idea. Metaphysics is not the study of the supernatural. It is the study of the most general structure of reality, whatever that structure turns out to be.

1.2 The ladder of "why"

Children discover the central problem of metaphysics by themselves. Ask "why?" after every answer and you climb a ladder. The glass broke because it fell. It fell because of gravity. Gravity works as it does because of the laws of general relativity. Then you ask why *those* laws hold, and finally why there is a lawful physical world at all.

Every rung but the last is answered from inside the physical order

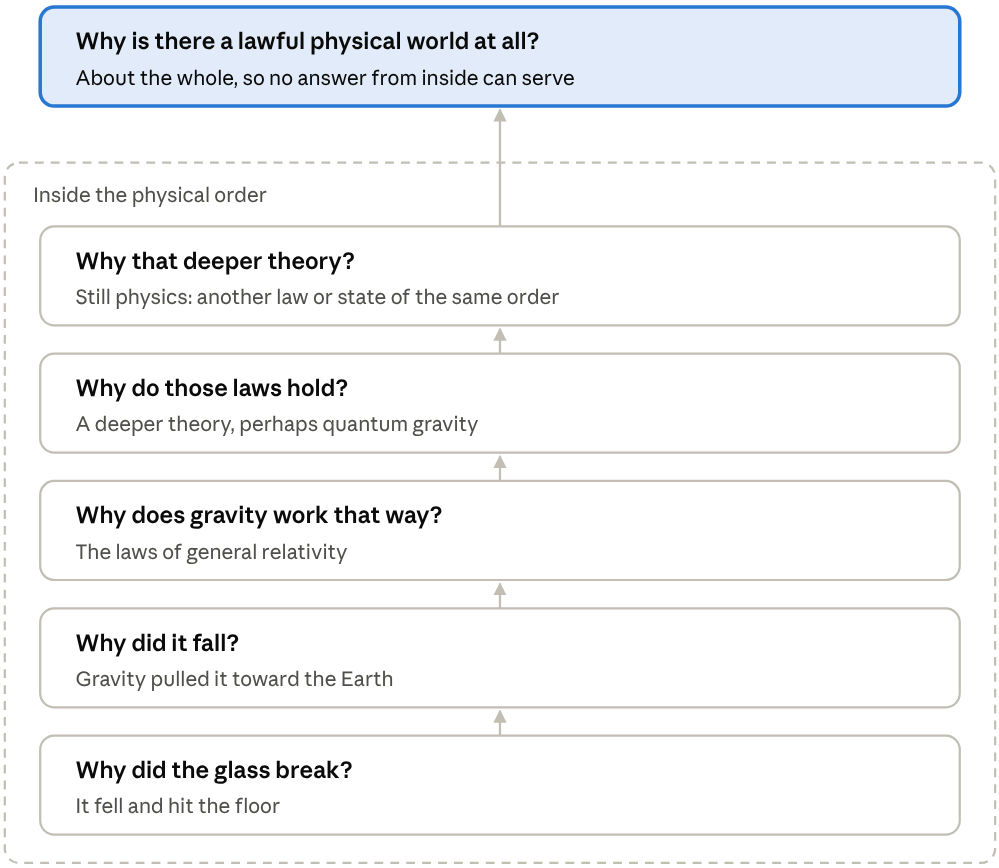


Figure 1.1 · The ladder of “why” questions

Each rung explains the one below by pointing to something more general. The lower rungs are answered from *inside* the physical world: one physical thing or law explains another. The top rung is different in kind. It asks about the physical world as a whole, so an answer drawn from inside that world would assume the very thing in question.

This book is about the top rung. Its aim is not to answer the question but to state it so exactly that every answer can be compared with every other.

1.3 Why the top question is hard

There are four separate difficulties. Each one becomes the subject of a later chapter.

1. **Every chain of explanation must end somehow.** It can go on forever, loop back on itself, or stop. Chapter 4 studies the endings.
2. **No experiment can stand outside the world.** Every measurement happens inside the physical order. Chapter 5 says exactly what “inside” means, and what physics can and cannot contribute.

3. **Language strains.** “Nothing” is not a very empty something. “The universe as a whole” may not behave like its parts. Chapter 6 examines what the question takes for granted.
4. **People mean different things by “God.”** A Christian theologian, a Hindu Vedāntin, a deist and a Stoic can all say “God exists” and mean four different claims. An atheist may be denying a fifth. Philosophers call this *equivocation*: one word, several meanings, and an argument that slides between them. Chapter 7 removes it.

1.4 How metaphysics is done, and who doubts it

Metaphysicians cannot experiment on the whole, so they use four methods.

- **Conceptual analysis:** clarify what a concept requires.
- **Inference to the best explanation:** prefer the theory that explains the most with the least.
- **Theoretical virtues:** weigh simplicity, scope, coherence and fit with science.
- **Thought experiments:** imagine cases that test a definition.

The subject has famous critics, and they matter to this book because their positions become one of the four answers.

- **Hume** said that claims which are neither mathematics nor experience should be “committed to the flames” (→ A.12).
- **Kant** argued that reason contradicts itself when it reaches beyond possible experience (→ A.13).
- **The logical positivists** of the 1920s and 1930s called such claims meaningless (→ A.21).

Metaphysics revived after the 1950s through Quine, Kripke and Lewis (→ A.22). The framework built here does not assume the critics are wrong. It gives them a place: the **Refused** verdict of Chapter 9.

Built so far. *There is a question at the top of the ladder: why is there a lawful physical world at all?* Every word in that sentence still needs to be made exact.

Chapter 2 — Two facts about the world

The top question asks why there is “a lawful physical world.” This chapter finds out how many separate facts that phrase contains. The answer is two, and the chapter defends that number.

2.1 Start with what everyone grants

A shared question must start from facts that every party accepts. Set aside everything contested: souls, gods, numbers, values, purposes. Two facts remain that a theist, an atheist, a Buddhist and a physicist all grant, at least at the level of everyday life.

1. **There are physical things.** Tables, stars, electrons.
2. **They behave regularly.** Dropped things fall. Copper conducts. The same experiment gives the same result tomorrow.

Call the first **G-existence** and the second **G-law**. Stated exactly, G-existence is "that there is a physically structured domain at all, rather than no such domain," and G-law is "that this domain is lawful — regular and structured, not a heap of disconnected happenings."

2.2 Are these really two facts?

Try to imagine each without the other.

- **Things without regularity.** A world of stuff that behaves at random from moment to moment seems conceivable: a "heap of disconnected happenings." So the first fact does not obviously bring the second with it.
- **Regularity without things.** Laws with nothing at all to govern are much harder to conceive. So the second fact seems to need the first.

That suggests G-law is a *further* fact on top of G-existence. Whether it really is depends on what a law of nature is, and philosophers disagree about that. The next two sections lay out the options, because the choice affects which answers to the top question are available.

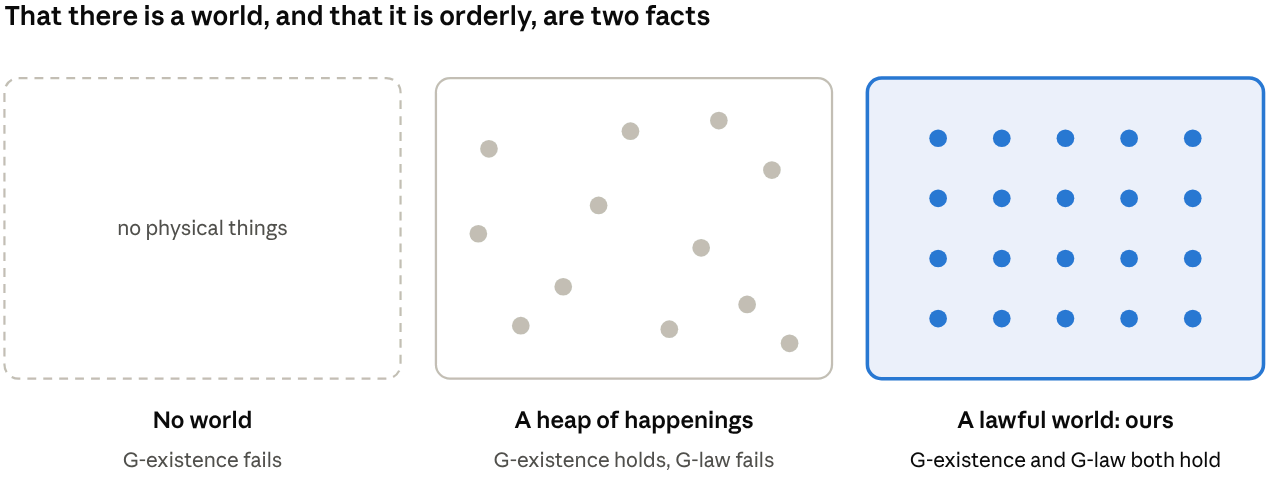


Figure 2.1 · Two facts, three possible worlds

2.3 What a law of nature is

"All copper conducts electricity" is a law. "All the coins in my pocket are copper" is an accident. Both are true generalizations. The difference shows in **counterfactuals**: if this wire were copper, it *would* conduct, but if this coin were in my pocket, it would not thereby become copper. A classic pair makes the point.

- "No sphere of uranium-235 is a mile wide" is law-like. Such a sphere would explode long before it reached that size.
- "No sphere of gold is a mile wide" is true only by accident.

There are four main theories of what makes the difference, and two theistic ones.

1. **Humean regularity, or the "best system" (David Lewis).** There is only the "mosaic" of particular facts: what is where, when. The laws are the axioms of the best summary of the mosaic, the one that best balances simplicity and strength. Laws describe. They do not push.
2. **Governing laws (Armstrong, Dretske, Tooley).** A law is a relation of *necessitation* between properties: being F necessitates being G. Laws are something over and above the mosaic, and they explain it.
3. **Powers, or dispositional essentialism (Bird, Ellis, Mumford).** Properties are *powers*. Having negative charge just *is* being disposed to repel other negative charges. The laws flow from the natures of the properties.
4. **Primitivism (Maudlin).** Laws are fundamental and cannot be reduced to anything else.
5. **Divine decree (Descartes, Boyle; recently Foster).** The laws are what God commands of matter.
6. **Occasionalism (al-Ghazālī, → A.7).** There are no necessary connections in nature. The regularities are God's habitual way of acting.

Figure 2.2 places them on two axes. Do laws merely *summarize*, or do they *do work*? Is their source *inside* nature or *beyond* it?

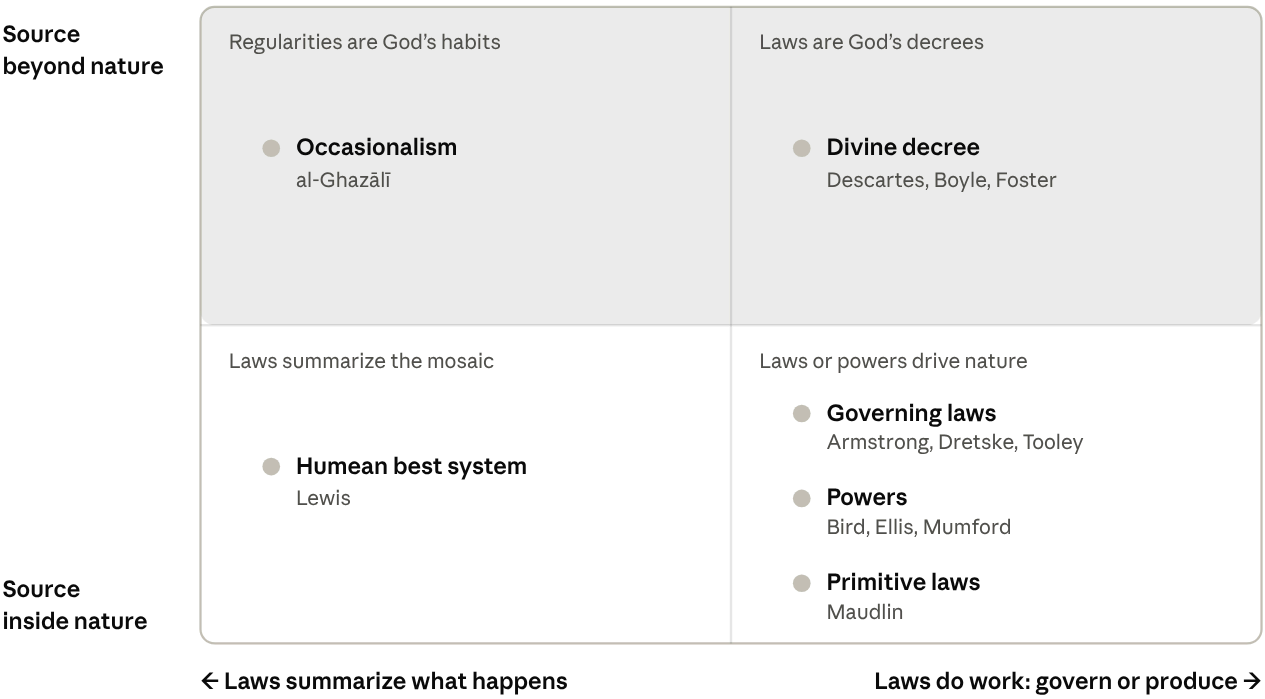


Figure 2.2 · Theories of what a law of nature is

On the theories in the shaded upper half, lawfulness has a source beyond nature by construction. The lower half leaves that open.

> **Ladder — what is a law of nature?**
>
> **Rung 1.** Things in nature always behave in the same ways: dropped things fall, ice melts when warm. A “law of nature” is a name for one of those always-the-same ways.
>
> **Rung 2.** Scientists state laws as rules, like F = ma. Is such a rule only a short description of what always happens, or something that *makes* things happen that way? A traffic law makes people drive on one side of the road. A weather report only describes. Which is a law of nature more like?
>
> **Rung 3.** Humeans take the second view: laws supervene on the mosaic, and the best-system account says which regularities count as laws. Anti-Humeans (governing, powers and primitivist views) say laws or powers come first and *explain* the regularities. The main objection to Humeanism is that if laws only describe, the universe’s pervasive regularity looks like a colossal coincidence.
>
> **Rung 4.** Open problems include the *inference problem* for governing views (how does a necessitation relation between universals entail that all Fs are Gs? van Fraassen 1989), whether powers views can handle conservation laws and symmetries, and whether a Humean can explain anything by citing a law that is itself a summary of what is to be explained. For this book, the one issue that matters is whether lawfulness is a fact *over and above* the existence of the mosaic.

2.4 What each theory implies

Theory of laws	Is lawfulness a fact beyond existence?	What follows for the top question
Humean best system	Only in part. Lawfulness is the mosaic’s orderliness.	Whatever fixes the whole mosaic fixes the laws. The two facts can still come apart if existence is unexplained and its orderliness is explained separately.
Governing laws	Yes. Necessitation relations are extra items.	The stuff and the laws could have different sources.

Theory of laws	Is lawfulness a fact beyond existence?	What follows for the top question
Powers	No. Laws follow from which properties exist.	Whatever explains the propertied things explains the laws. An orderer of pre-existing stuff is still possible.
Primitivism	Yes. Laws are basic.	The two facts are fully distinct.
Divine decree, occasionalism	Yes, and it has a source.	The law-side is explained by construction.

Key idea. G-law is a separate fact from G-existence on most theories of laws, and partly separate on the rest. The framework therefore treats them as two sides, and lets each theory of laws say how far apart they can come.

2.5 Each side has a general and a specific part

“Why is there law at all?” differs from “why *these* laws?” Derek Parfit called the second question “Why this?” (→ A.23). The same split applies to existence: “why anything?” differs from “why this inventory of things, in this initial state?”

Several constants of physics appear to lie in narrow ranges that permit chemistry and life. This is the **fine-tuning** observation. Its standard explanations are design, a multiverse with a selection effect, a deeper law, or luck.

The framework folds the specific questions into the two sides.

- **G-existence** covers *that* there is a physical order, and its basic inventory and initial state.
- **G-law** covers *that* the order is lawful, and which laws hold.

An explanation that covers only the general part of a side is marked “generic only” (Chapter 11).

2.6 Why two sides, and not three or five?

A critic will ask why consciousness, or value, or mathematics is not a third side. The answer needs a rule for what counts as a side. A fact is a **side** of the physical order only if it meets three conditions.

1. **Common ground.** Every party to the debate grants the fact.
2. **Global.** It is a fact about the order as a whole, not about some part of it.

3. **Distinct.** On at least one live theory, it is not already settled by the other sides.

Candidate side	Common ground?	Global?	Distinct?	Result
There is a physical order	yes	yes	yes	**Side: G-existence**
It is lawful	yes	yes	yes, on most theories of laws	**Side: G-law**
It has these particular laws and constants	yes	yes	no: it is the specific form of G-law	Folded into G-law
It can be described by mathematics	yes	yes	no: an aspect of lawfulness	Folded into G-law
It contains conscious beings	disputed: physicalists say consciousness is part of the physical order, dualists say it is outside it	no	—	Not a side of the *physical* order
It contains objective value	disputed	—	—	Not a side

Consciousness and value fail the first condition. People do not agree that they are features of the physical order at all. A reader who wants them covered can use the wider roles of Chapter 11, which change the *domain* to be explained without adding sides to this one.

Built so far. *Why is there a physical order (G-existence), and why is it lawful (G-law)?* Two sides, each with a general and a specific part. The next chapter examines the word “why.”

Chapter 3 — Kinds of “because”

The question so far asks *why* there is a physical order and *why* it is lawful. An answer will have the form “because of X.” This chapter asks what “because” can mean, and builds the one relation the framework needs.

3.1 What an explanation is

An explanation has two parts. The *explanandum* is what needs explaining. The *explanans* is what does the explaining. Aristotle noticed that "why?" has several kinds of answer, later called his four causes (→ A.4). For a bronze statue:

- the **material** cause is the bronze it is made of;
- the **formal** cause is its shape;
- the **efficient** cause is the sculptor who made it;
- the **final** cause is what it is for.

Explanation is also often **contrastive**: we explain why this happened *rather than* that. G-existence contrasts a physical order with none. G-law contrasts an orderly world with a chaotic one.

3.2 Two modern kinds of "because"

Compare two sentences.

- The window broke *because* the ball hit it.
- The ball is red *because* it is scarlet.

The first "because" is **causal**. One event produces a later one. The second is not causal. Nothing happens first. Being scarlet simply *is one way* of being red, and the ball is red *in virtue of* being scarlet. Since about 2009, metaphysicians have called the second relation **grounding** (→ A.17). Other standard examples: a set exists in virtue of its members, and a statue has its shape in virtue of how its clay is arranged.

On most accounts grounding is:

- **irreflexive:** nothing grounds itself;
- **asymmetric:** if A grounds B, B does not ground A;
- **transitive:** if A grounds B and B grounds C, then A grounds C;
- **necessitating:** a full ground guarantees what it grounds.

Kit Fine, Jonathan Schaffer and Gideon Rosen developed the theory. Jessica Wilson (2014) argued that "grounding" is an umbrella over many specific relations, and that the specific ones do the real work.

> **Ladder — grounding**
>
> **Rung 1.** Some "because"s are about what happened first: the tower fell because I kicked it. Other "because"s are about what something *is*: the tower is tall because it has lots of blocks. The second kind has no "first."
>
> **Rung 2.** Causation links events in time. Grounding links facts by what makes them so: "this shape is a square in virtue of having four equal sides and four right angles." A ground does not come before in time. It is what the fact holds in virtue of.
>
> **Rung 3.** Grounding is usually treated as a strict partial order. It is also *hyperintensional*: it separates facts that always hold together. The set {Socrates} exists in virtue of Socrates, not the other way round, though each exists exactly when the other does. Full grounds necessitate what they ground.
>
> **Rung 4.** Open issues: whether grounding is one relation (Fine, Schaffer) or a family (Wilson); whether it is well-founded (Schaffer says yes, Bliss doubts it); whether it must necessitate; whether it relates facts or things; and how it relates to causation (Schaffer 2016 models it on causal structural equations). The necessitation issue matters most here, as the next section shows.

3.3 Neither kind is neutral enough

Which "because" should the top question use? Each choice, taken alone, quietly favours some answers over others.

If "because" means strict grounding, two things go wrong.

- *Necessitation.* Classical theism holds that God created *freely* and could have refrained. A free choice does not guarantee its result, so on the strict reading a free creator would not count as a ground. Theism could not even state its own answer.
- *Constitution.* Paradigm grounds are constituents or more basic aspects of what they ground. A creator is not a constituent of the world. The strict reading would favour views on which God and world are one.

If "because" means causation, the opposite bias appears.

- Causation normally links events in time. But Augustine argued that time itself came with the world (→ A.6), and Aquinas argued that even a world with no beginning would depend on God (→ A.8). The ground sought is therefore not an earlier event in time.
- Views on which the world flows from, expresses, or is required by its source would be shut out, because those are not causes in the ordinary sense.

Some recent philosophers have argued that God should be understood as *grounding* the world, not causing it (Pearce 2017; Bohn 2018). Others keep creation causal. A shared vocabulary cannot take a side in that dispute.

3.4 Ground*: one relation, four species

The solution takes Wilson's point constructively. Define a general relation, and require every answer to say which specific kind it means.

Definition. x **grounds*** a fact f if and only if three things hold.

1. **Explanation.** f obtains at least partly *because of* x. On the view being scored, x figures in a correct, non-trivial explanation of f.
2. **Asymmetry.** f does not in the same sense explain x.
3. **Species.** The explanation is of at least one of the four kinds below.

Two things are deliberately *not* required. x need not guarantee f, so a free creator counts. And x need not come earlier in time than f, or be in time at all.

Species	Tag	How x explains the order	Examples
Productive	p	x brings about, emanates or conserves it	Abrahamic creation; Plotinus's emanation
Constitutive	c	the order is an aspect, mode, body or appearance of x	Spinoza's substance; Advaita's Brahman
Nomic	n	x fixes or legislates the laws	Divine decree; Plato's Forms; Zhu Xi's *li*
Axiological	a	x's value requires or attracts the order	Plato's Good; Aristotle's mover as final cause; axiarchism

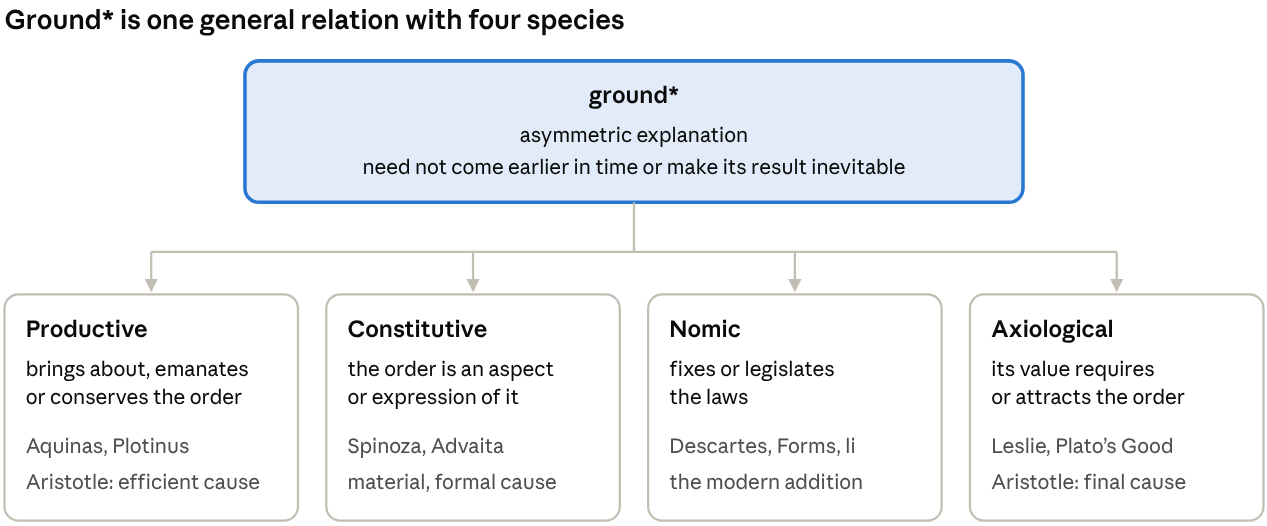


Figure 3.1 · Four species of ground*

The four species are descendants of Aristotle's four causes. Productive is his efficient cause. Constitutive combines his material and formal causes. Axiological is his final cause. Nomic is the modern addition, made necessary once "laws of nature" became a category of their own.

Is this too generous? A general relation might seem to let anything count. Three things prevent that. The relation must be asymmetric. The explainer must be outside the physical order, in a sense Chapter 5 makes exact. And the framework records what a view *claims*; it does not certify that the claimed explanation succeeds. A critic who thinks axiological explanation is a confusion can say so. That is an objection to an answer, not to the question.

One technical note. Tests of the form "if x had not existed, f would not have obtained" fail for necessary beings, because a conditional with an impossible antecedent is trivially true on the standard account. That is why ground* is defined through explanation, not through counterfactuals.

> **Ladder — ground***
>
> **Rung 1.** There are different ways one thing can be the reason for another. A baker makes a cake. A cake is made *of* flour and eggs. A recipe says how the cake should be. And a birthday is the reason there is a cake at all. Any of these kinds of reason is allowed.
>
> **Rung 2.** "Ground*" means "the thing that explains why the world is there and orderly." It could be a maker, something the world is an expression of, a rule-giver, or a value. It does not have to be earlier in time, and it does not have to force the world to exist.
>
> **Rung 3.** Ground* is a determinable whose determinates are productive, constitutive, nomic and axiological dependence. It is asymmetric and explanatory. It is broader than Fine-style grounding, which necessitates, and narrower than "any explanation," because it requires a named species and, in the role, an external explainer.
>
> **Rung 4.** This is a pluralist reading of "big-G" Grounding in Wilson's sense. Open problems: whether a constitutive ground can be genuinely outside what it constitutes (Chapter 5 answers with the relocation test); whether axiological explanation is a real species (Leslie and Rescher say yes); and how the relation behaves across species under transitivity (Chapter 10 handles chains with a terminal clause).

Built so far. *What, if anything, grounds* the fact that there is a physical order, and the fact that it is lawful?* The next chapter asks how such a chain of explanation can end.

Chapter 4 — Where explanations stop

A chain of "because" answers must end somehow. This chapter lists every way it can end. The list matters because each ending becomes a recognizable answer to the top question.

4.1 Agrippa's trilemma

The ancient skeptic Agrippa noticed that any chain of reasons has only three possible shapes (→ A.2).

- It goes on forever: a **regress**.
- It loops back on itself: a **circle**.
- It **stops**.

A stopping point, in turn, is one of two kinds. It is a **brute fact**, true for no reason at all. Or it is **self-explaining**: something whose existence is accounted for by its own nature, the usual candidate being a *necessary* being.

> **Ladder — Agrippa's trilemma**
>
> **Rung 1.** If you keep asking "why?", only three things can happen. You keep going forever, you come back to an answer you already gave, or you stop at an answer that has no answer behind it.
>
> **Rung 2.** A reason needs support, and its support needs support. The chain is infinite, circular, or finite and ending in something unsupported. Each option looks unsatisfying at first, and philosophers still disagree about which is least bad.
>
> **Rung 3.** In the theory of knowledge the trilemma drives the choice between infinitism, coherentism and foundationalism. In metaphysics it drives the choice between an infinite regress of explainers, self-explanation, and a brute or necessary terminus. Leibniz argued that even an infinite regress leaves the *whole series* unexplained (→ A.10).
>
> **Rung 4.** The trilemma is exhaustive only for *linear* chains. Some philosophers argue that explanation can be holistic, with facts supporting each other in a web. Others deny that dependence must be well-founded. The framework built here does not rule these out. A view that rejects the chain model for the world as a whole is recorded under Refused (Chapter 9).

4.2 Necessity and contingency

A truth is **necessary** if it could not have been false, and **contingent** if it could have been. Philosophers separate three grades.

- **Logical:** "all bachelors are unmarried."
- **Metaphysical:** "water is H_2O," according to Kripke (→ A.20).
- **Nomic, or physical:** "nothing travels faster than light," given our laws.

A useful picture is *possible worlds*: complete ways reality could have been. Necessary truths hold in all of them. Contingent truths hold in some.

The physical world appears contingent. It seems it could have failed to exist, or had different laws. Many arguments for an outside ground start there. Some naturalists reply that the physical world, or its first state, is itself necessary.

> **Ladder — necessity**
>
> **Rung 1.** Some things couldn't be any other way: a triangle always has three sides. Other things just happen to be true: your shirt is blue today.
>
> **Rung 2.** Imagine every complete way things could have gone. "Necessary" means true in every one. "Contingent" means true in some and not others.
>
> **Rung 3.** The grades differ in which worlds count: all logically possible worlds, all metaphysically possible ones, or only those with our laws. A *necessary being* exists in every metaphysically possible world. The argument from contingency moves from a contingent world to a necessary being.
>
> **Rung 4.** Open questions: whether metaphysical necessity is grounded in essence (Fine 1994); whether a necessary being can explain a contingent world without making the world necessary too (§4.3); and whether a "necessary initial state" of the universe is coherent (Oppy 2013).

4.3 Brute facts and the principle of sufficient reason

The **principle of sufficient reason** (PSR), stated most famously by Leibniz (→ A.10), says there are no brute facts. Every fact has a reason why it is so and not otherwise.

The PSR drives most arguments for an outside ground. If every fact has an explanation, so does the fact that the physical world exists. An infinite chain of physical causes explains each member, but not the chain. So the explanation must lie outside the chain.

The most famous objection is **modal collapse** (van Inwagen 1983). Join all contingent truths into one giant fact. By the PSR it has an explanation. A contingent explainer would be part of what it explains. A necessary explainer that *entails* it would make it necessary. So the PSR seems to make everything necessary. Defenders reply that an explanation need not entail what it explains. A free choice can explain without necessitating (Pruss 2006).

> **Ladder — the PSR and modal collapse**
>
> **Rung 1.** The rule “everything has a reason” sounds right. But then “everything that happens” needs a reason too, and that reason can’t be one of the things that happens, or it would be explaining itself.
>
> **Rung 2.** Leibniz said the world as a whole must have a reason outside it: a being that exists necessarily. Critics reply that if that reason *forces* the world to be as it is, nothing could have been different, and that seems false.
>
> **Rung 3.** Let p be the conjunction of all contingent truths. If the PSR holds, some q explains p. If q is contingent, it is a conjunct of p, and the explanation is circular. If q is necessary and entails p, then p is necessary. Responses: deny that explanation requires entailment, restrict the PSR, or accept a brute fact.
>
> **Rung 4.** Pruss defends a PSR for contingent facts with free choice as a non-necessitating explanation. Della Rocca (2010) defends an unrestricted PSR and accepts its Spinozist consequences. Dasgupta (2016) suggests some facts are “autonomous”: neither explained nor in need of explanation. The framework treats these as routes to different answers and takes no side.

4.4 The endings, applied to the top question

Put the pieces together. When someone asks what grounds* the physical order, the chain of explanation can end in five ways.

Ending	What it says about the physical order	Becomes, in Part II
Brute	It exists and is lawful for no reason	Empty (brute)
Necessary	It could not have failed to exist	Empty (necessary)
Regress inside	Each state is explained by an earlier one, forever, and that is all	Empty (regress)
Outside ground	Something that is not part of it grounds* it	Split or Filled
No chain	The demand for an explanation of the whole is rejected	Refused

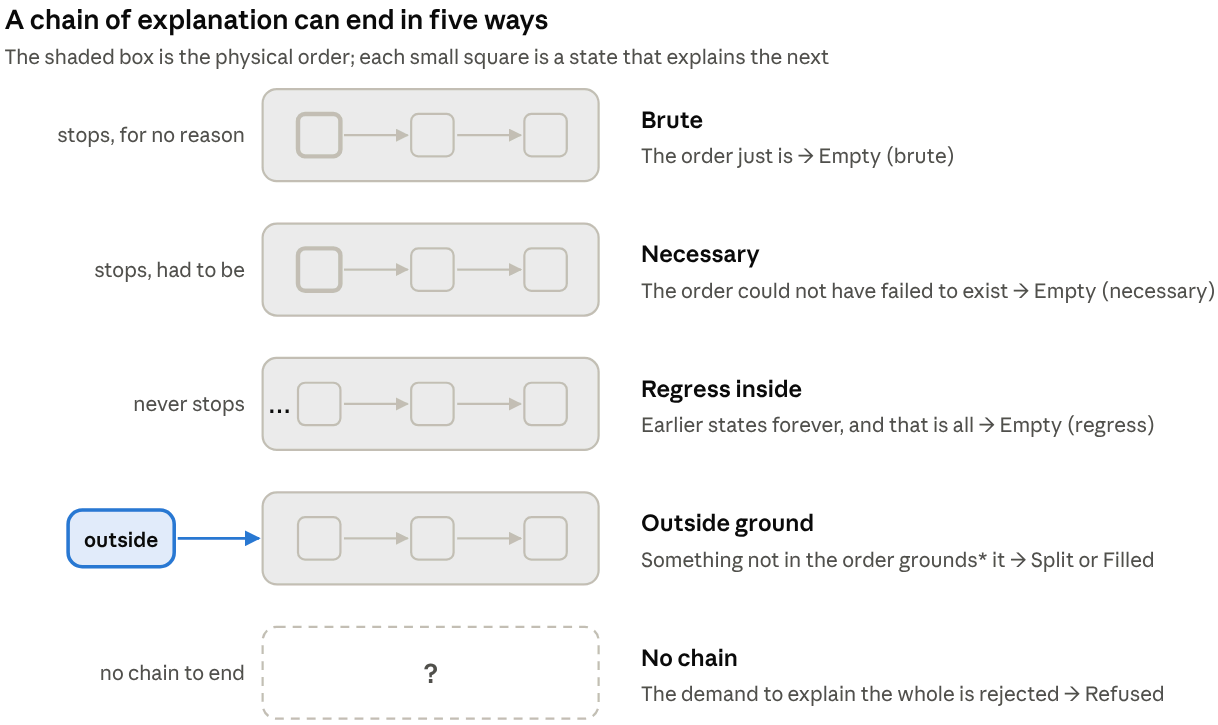


Figure 4.1 · Five ways a chain of explanation can end

An outside ground must itself end in one of the first two ways, as brute or as necessary. That is why the old exchange — "God is the necessary being"; "then why can't the world be?" — never goes away. Both sides are choosing where to place the same kind of stopping point.

Key idea. Nobody escapes the trilemma. Every answer to the top question is a decision about where explanation stops, and whether it stops inside the physical order or outside it.

Built so far. *What, if anything* outside the physical order *grounds* the fact that there is such an order, and the fact that it is lawful?* The word "outside" now carries the weight. The next chapter defines it.

Chapter 5 — Inside and outside

The question now asks for something *outside* the physical order. This chapter says what the physical order is, what counts as inside it, and why the best theories of physics, however deep, are inside by their nature. The framework's word for inside is **internal**.

5.1 What a physical explanation looks like

Every explanation in physics has the same form: **a state of the world, plus laws, gives another state of the world.** The early universe plus the laws of gravity gives galaxies. A quantum state plus the Schrödinger equation gives a later quantum state.

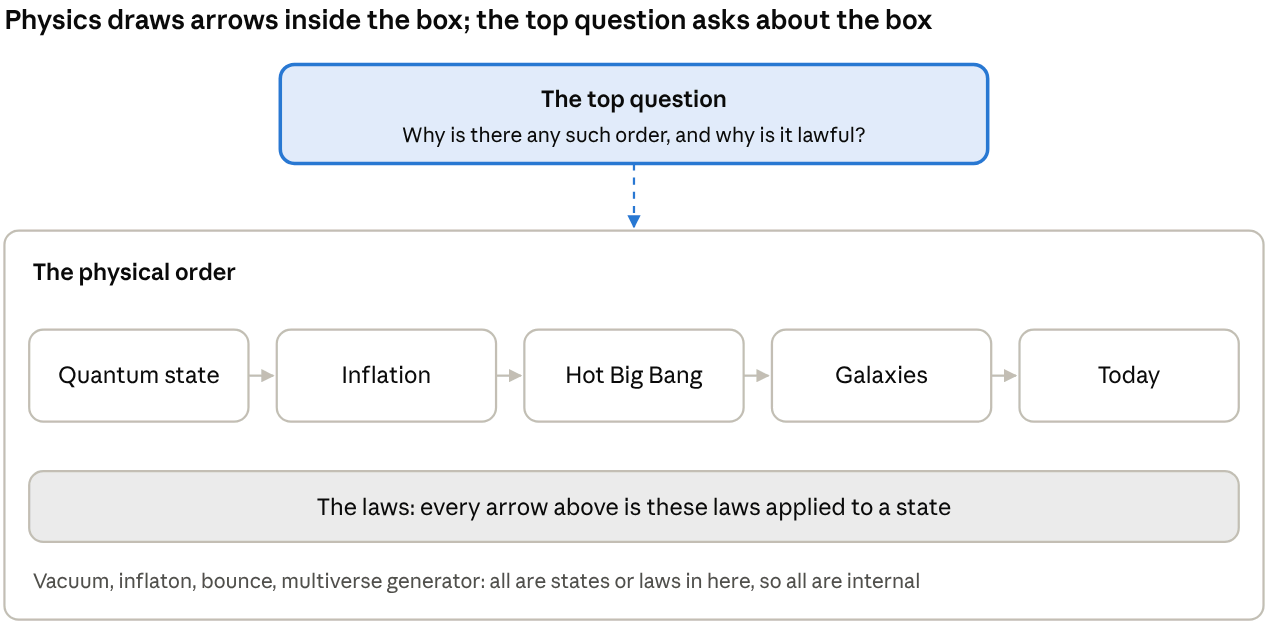


Figure 5.1 · Every physical explanation starts from something physical

Every arrow physics can draw runs between things inside the box. The top question is about the box.

5.2 A short tour of cosmology

The framework counts the Big Bang, inflation, bounce cosmologies, quantum gravity and multiverse generators as internal. Here is what each one is.

The Big Bang. In 1927 Georges Lemaître, a Belgian physicist and priest, showed that Einstein's equations allow an expanding universe (→ A.18). Edwin Hubble's observations (1929) supported expansion. The discovery of the cosmic microwave background by Penzias and Wilson (1965) confirmed that the early universe was hot and dense. The theory says the observable universe was in that state about 13.8 billion years ago. It does **not** say the universe came from nothing. Run the equations backward and they break down at a **singularity**. That marks where the theory stops being reliable, not a proven first moment. Lemaître himself objected when Pope Pius XII, in a 1951 address, treated the theory as support for creation.

Singularity theorems. Hawking and Penrose proved that, under general conditions, Einstein's theory predicts singularities. In 2003 Borde, Guth and Vilenkin proved that any spacetime that has on average been expanding is *past-incomplete*. Some read this as evidence of an absolute beginning. Most cosmologists read it more narrowly: classical expanding spacetime has a past boundary, and what lies beyond it awaits a quantum theory of gravity.

Inflation and the multiverse. Alan Guth (1981) proposed a brief burst of exponential expansion in the very early universe. In many versions inflation never ends everywhere. It stops in patches, which become "bubble universes," and continues elsewhere. The result is a **multiverse** in which the constants may differ from bubble to bubble.

Quantum cosmology. Alexander Vilenkin (1982) described a universe "tunnelling from nothing." James Hartle and Stephen Hawking (1983) proposed a *no-boundary* state in which time becomes space-like near the beginning, so there is no edge. Asking what came before is then like asking what lies north of the North Pole.

Bounces and cycles. Some models replace the Big Bang with a bounce from an earlier contracting phase, or with endless cycles (Steinhardt and Turok; Penrose 2010). On these models the physical order may have no first moment.

5.3 Why all of this is inside

In 2012 the physicist Lawrence Krauss published *A Universe from Nothing*, presenting quantum cosmology as an answer to why there is something rather than nothing. The physicist-philosopher David Albert replied in a review. The "nothing" in these models is a quantum vacuum, or a state with no classical spacetime. It is still a physical state, governed by the laws of quantum mechanics. The models show how one physical situation gives rise to another. They do not show why there are physical situations and laws at all.

The same holds for each item in the tour. The multiverse generator is a physical field obeying physical laws. A bounce is one state of the world following from another. Each is an arrow inside the box.

A beginningless universe does not change this. Aquinas and Avicenna held that a world with no first moment would still depend on a ground (→ A.7, A.8). Aristotle held that an eternal world still needs a source of its motion (→ A.4). Whether the physical order has a first moment is a question for physics. Whether it has an outside ground is a different question.

> **Ladder — can physics get a universe from nothing?**
>
> **Rung 1.** Scientists sometimes say a universe can pop out of "nothing." But their "nothing" still has rules for how things can pop. Real nothing wouldn't even have rules.
>
> **Rung 2.** A vacuum in physics is not empty. It is the lowest-energy state of fields, and it is full of activity. A universe arising from a vacuum is a change from one physical situation to another. That answers "how did our universe start?", not "why is there anything at all?"
>
> **Rung 3.** Vilenkin's and Hartle and Hawking's proposals assign quantum amplitudes to possible spatial geometries using a wave function that obeys the Wheeler–DeWitt equation. Both presuppose a space of states and quantum laws. "There is a lawful physical order" is assumed, not derived. This is Albert's point against Krauss.
>
> **Rung 4.** A subtler proposal: perhaps a final physics will be *self-explaining*, its laws the only consistent possibility. If so, the physical order would be necessary. That would not put anything outside the order. It would be one of the endings of Chapter 4, and Part II scores it as Empty (necessary). Physics cannot occupy the outside position. It could, in principle, show that the position is idle.

None of this makes physics irrelevant. Physics can discover whether the order had a first state, whether its laws are unique or fine-tuned, and whether there are many universes. Those findings change which answers are plausible. They do not change the question.

5.4 What "physical" means

Saying what is "inside the physical order" requires saying what "physical" means, and that is harder than it looks. Carl Hempel (1969) posed a dilemma.

- **If "physical" means "what current physics describes,"** the definition is almost certainly wrong, since current physics is incomplete.
- **If it means "what a final, ideal physics would describe,"** the definition is empty, since nobody knows what that physics will contain.

The way out, adapted from Daniel Stoljar's *object-based* approach, is to define the physical by pointing, not by theory.

Definition. Start with paradigm physical objects: this table, that star, this electron. The **maximal physical order, P***, is everything connected to the paradigms by a chain of links. A link is one of three kinds.

1. **Spatiotemporal relation.** Being at some distance or time from.
2. **Lawful interaction.** Two items that affect each other, both subject to laws.
3. **Lawful realization.** One item is made up of, or runs on, the law-governed states of another, as a wave is made up of water or a program runs on a computer.

The third link is new in this edition, and the simulation case shows why it is needed. Suppose our universe is a simulation running on a computer in some "base" world. A simulated table is not at any distance from the computer, and it does not push or pull on it. It is *realized by* the computer's states. With the third link, the computer and its whole world belong to P*.

A one-way ground* relation, such as creation, is **not** a link. It is not governed by laws that both sides are subject to.

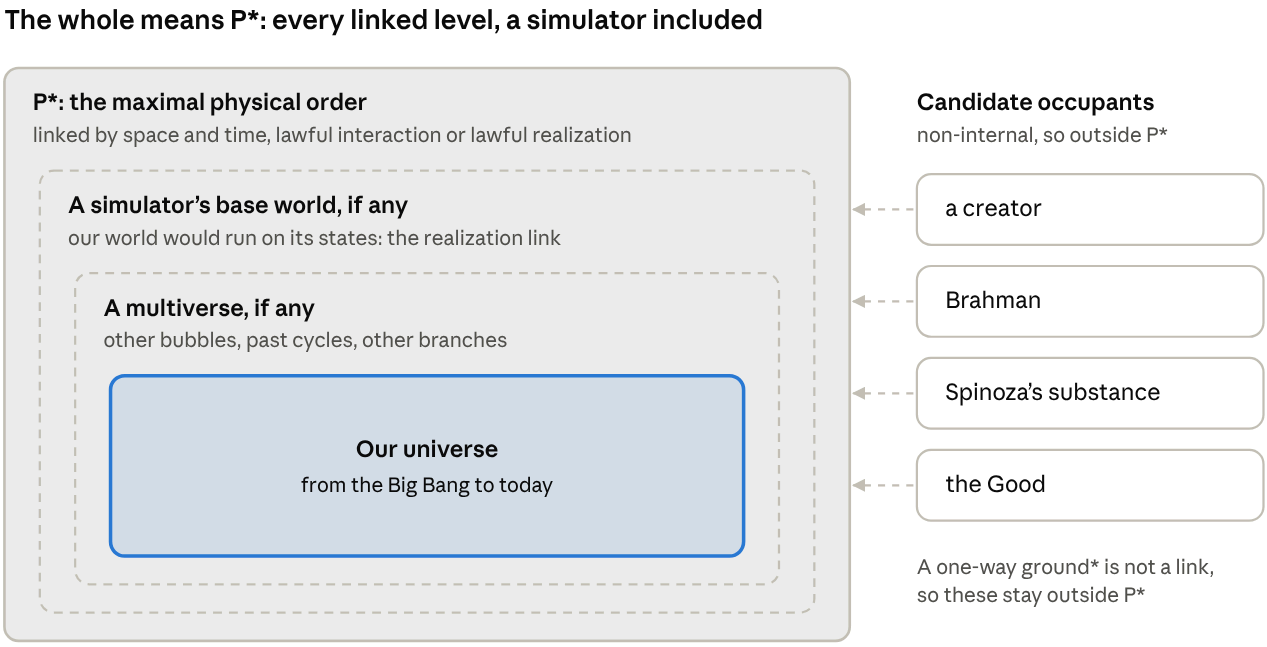


Figure 5.2 · Nested orders and the maximal physical order P*

5.5 The test for "internal"

Definition. An item x is **internal** if and only if at least one of three conditions holds.

- **(a) Member.** x is in P*.
- **(b) Relocation.** x is itself a domain of things or states that behave regularly, so that "why is there x, and why is it regular?" is just the two-sided question over again. Citing x relocates the question without addressing it.

- **(c) Identity.** x is the same thing as P*, or an aspect of P*, or part of what was to be explained. “The laws” cannot ground* lawfulness, because the laws are the lawfulness.

Otherwise x is **non-internal**.

The intrinsic-description rule. Apply the test to what the view says x *is in itself*, not to a label that picks x out by its relation to the world. “The creator of the universe” mentions the universe, but that does not make a creator internal. What matters is what the view says the creator is: for example, a necessary, immaterial mind.

What the relocation test does not say. It does not say that an outside ground raises no further questions. “Why does God exist?” is a fair question, and Chapter 4 showed that it ends in brute fact or necessity like every chain. The test is narrower. It asks whether the *same* question recurs: the question why there is a regular domain of states. For a simulator’s computer it plainly does. For a being that a view describes as simple, changeless and not subject to any laws, it does not, whatever else may be asked about such a being.

Candidate	(a)	(b)	(c)	Result
The Big Bang, or a first state	yes			internal
An inflaton field, a multiverse generator	yes			internal
A quantum vacuum, a no-boundary wave function		yes		internal
A simulator’s computer and its world	yes	yes		internal
The laws of nature themselves			yes	internal
Universals that exist only in their instances (Armstrong)			yes	internal
The Stoics’ fiery logos, a body within the cosmos	yes			internal
A mathematical structure said to *be* the universe (Tegmark)			yes	internal
A cosmic mind said to be the inner nature of physical stuff			yes	internal
A classical creator				non-internal
Plato’s Forms, existing apart from their instances				non-internal

Candidate	(a)	(b)	(c)	Result
Spinoza's substance, with attributes beyond the physical				non-internal
Advaita's Brahman				non-internal
An ethical requirement that there be a good world				non-internal

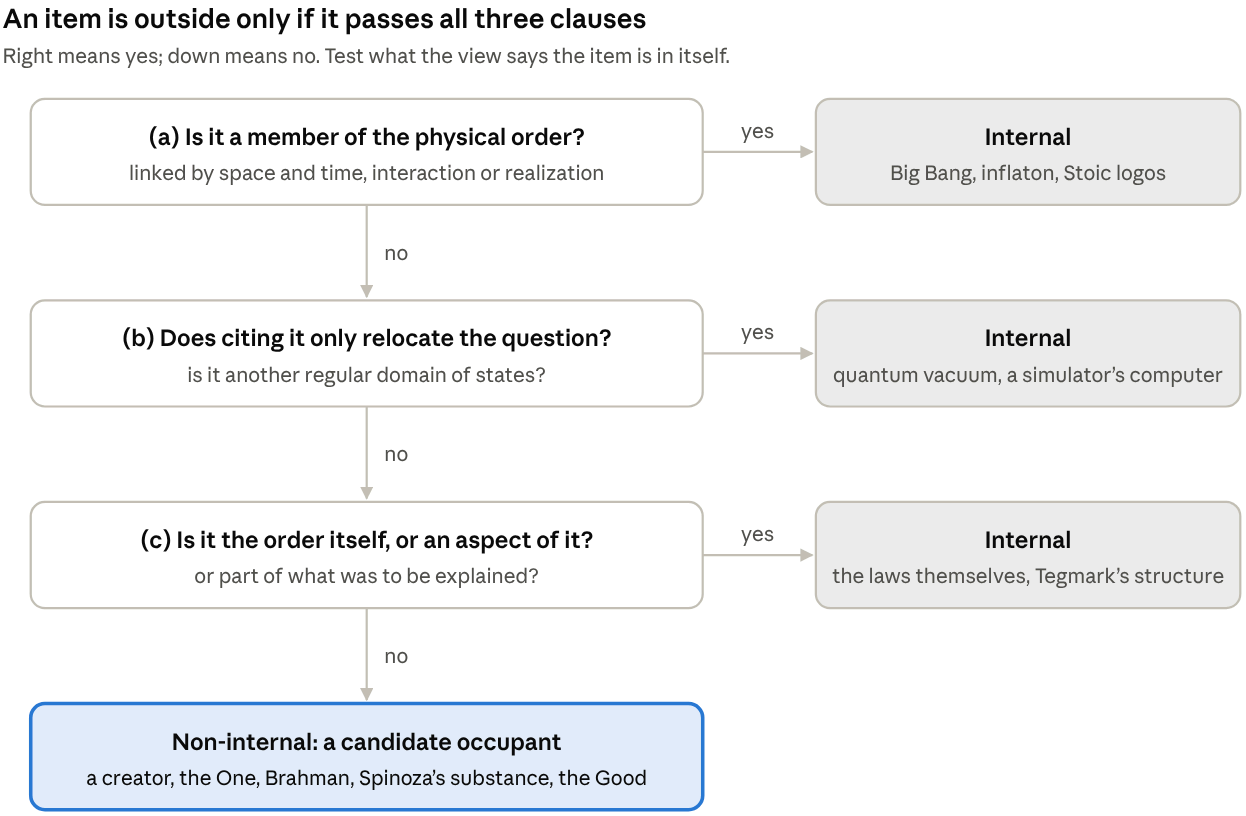


Figure 5.3 · The test for internal, as three questions

INTERNAL VS NON-INTERNAL

Something is internal if stating it already assumes a physical order: spacetime, laws, fields, particles, vacuum, history, or a generator that takes an ordered domain as input. Big Bang, inflation, bounce, quantum gravity, dark energy, and a multiverse generator are internal unless they can be stated without that assumption. An earlier stage of the same order is still internal.

Figure 5.4 · The internal test in one paragraph, as first stated

5.6 An honest boundary case

The hardest line to draw is between a *realizer* (internal, by the third link) and a *constitutive ground** (non-internal). Both stand "under" the physical order. The relocation test separates them. A computer is a law-governed system of states, so it relocates the question. Spinoza's substance, by Spinoza's account, is not a system of states obeying laws that could have been otherwise. It is the self-existent source of all laws.

Some views sit on the line. Bernardo Kastrup's idealism says the physical world is the outward appearance of a universal mind. If that mind has its own regular dynamics, a critic can say it relocates the question. A defender can reply that mind is self-existent in a way physical states are not. The framework does not settle this. It says exactly what the dispute is about, and Chapter 14 records the view as disputed.

Key idea. "Internal" is not a criticism. It is a job description. Physics explains the physical order from the inside. An item is outside only if it is not in the order, is not the order, and does not merely move the question somewhere else.

Built so far. *What, if anything, is a non-internal ground* of the fact that there is a physical order P*, and of the fact that it is lawful?* Two phrases remain: "the order as a whole," and the status of the word that will name the answer.

Chapter 6 — The whole, and what the question presupposes

The question asks about the physical order "as a whole." Two famous objections say there is something wrong with that phrase. This chapter shows that the question survives them, and in doing so finds exactly what the question takes for granted.

6.1 The fallacy of composition

What is true of each part need not be true of the whole. Each brick is small, but the wall is not. In a 1948 radio debate, Bertrand Russell told Frederick Copleston that every human being has a mother, but the human race does not have a mother (→ A.19). By analogy, each thing in the universe may have a cause without the universe having one.

Russell is right about the logic. From "each part has a ground" it does not follow that "the whole has a ground." But the framework makes no such inference. It only *asks* whether the whole has a ground, and it accepts "no" as an answer.

6.2 Twenty particles

In Part IX of Hume's *Dialogues concerning Natural Religion*, the character Cleanthes argues that if you have explained each of twenty particles of matter, it is unreasonable to ask for a further explanation of the twenty together. Uniting them into a whole is "an arbitrary act of the mind" (→ A.12).

The reply uses a tool from modern logic. George Boolos (1984) showed that we can talk about *some things, together* without supposing there is an extra object, the collection, that contains them. "The students surrounded the building" is not about any one student, and it does not need a mysterious entity called The Group. This is **plural logic**.

Applied here: G-existence is the plural fact that *there are some physical things*. That fact is not about any one of them. Explaining each particle by an earlier particle does not explain why there are any particles at all. So the question about the whole is a genuinely different question from the questions about the parts. That does not show it has an answer. It shows it is a real question.

Key idea. "The whole" in the definition means *the physical things, taken together*. No special super-object is assumed.

6.3 What a presupposition is

"Have you stopped cheating at cards?" cannot be answered well with "yes" or "no" if you never cheated. The question *presupposes* something false. P. F. Strawson (1950) argued that a statement with a failed presupposition, like "the King of France is bald," is neither true nor false (→ A.20). The right response is to reject the question.

A carefully built question should state its own presuppositions, so that anyone who rejects one can say which.

6.4 The three presuppositions of the top question

The question "what, if anything, is a non-internal ground* of the physical order as a whole?" takes three things for granted.

	Presupposition	What denying it sounds like
π_1 Totality	There are physical things, and it makes sense to speak of them all together	"There is no 'it' to ask about."

	Presupposition	What denying it sounds like
π_2 Explicability	It makes sense to ask what that totality depends on; dependence can apply to all of it, not only within it	"Explanation applies inside the world, not to the world."
π_3 Determinacy	Either something grounds* it or nothing does; the question has an answer, even if we cannot know it	"Neither 'grounded' nor 'ungrounded' is right."

Section 6.2 gave a reason to accept π_1. Chapters 3 and 4 gave reasons to accept π_2. Classical logic gives π_3. But serious thinkers have denied each one: Kant, Nāgārjuna, Carnap and the early Buddhist tradition among them. A vocabulary that aims to be shared cannot rule them out by definition. Part II gives them their own answer, **Refused**.

Built so far. *What, if anything, is a non-internal ground* of the physical order P* taken as a whole — of the fact that there are physical things, and of the fact that they are lawful?* The question presupposes totality, explicability and determinacy. Only one thing is missing: a name for whatever answers it.

Chapter 7 — Titles and roles: the definition assembled

The question is built. It needs a word for whatever answers it. The obvious word is "God," and the obvious danger is that the word brings a great deal with it. This chapter shows how to use the word safely, and states the finished definition.

7.1 Names, descriptions and titles

Three kinds of words pick things out.

- **Names.** "Venus" picks out a planet directly. Gottlob Frege noticed that "the Morning Star" and "the Evening Star" pick out that same planet while carrying different information (→ A.20). Saul Kripke argued that names are *rigid*: "Venus" names the same object in every possible situation.
- **Definite descriptions.** "The tallest mountain" picks out whatever fits. Bertrand Russell showed that "the present King of France is bald" is not about a ghostly non-existent king. It says that exactly one thing is King of France and that it is bald, which is false.
- **Titles.** "The President" names an office. Different people can hold it at different times, and it can be vacant.

In 1969 Nelson Pike argued that “God,” in classical theism, works as a title. It names the office of the being that is all-powerful, all-knowing and perfectly good. “Yahweh” or “Allah” names the claimed holder.

The difference matters at once. If “God” is a name, “God does not exist” is puzzling: what is the name naming? If it is a title, the sentence is plain. It says the office is vacant, as “there is no King of France” does.

> **Ladder — name versus title**
>
> **Rung 1.** “Serena” is a name: it points at one person. “Class captain” is a job: anyone could have it, and some years nobody does.
>
> **Rung 2.** You can argue about who the class captain is, or whether there is one, without first agreeing who it is. A job description lets people disagree about the answer while agreeing on the question.
>
> **Rung 3.** Names are rigid designators. Titles are non-rigid: they pick out whoever holds the office. Pike used this on the old puzzle of whether God can sin. If the title includes perfect goodness, “God sins” is necessarily false, while the individual who holds the office might still have the *power* to sin.
>
> **Rung 4.** How the divine names of the traditions actually function (as names, titles, or both) is disputed. This book avoids the dispute by stipulating a new, starred technical term and leaving every tradition’s own word alone.

7.2 Roles and occupants

Philosophers of mind made the idea of a role into a working tool. **Functionalism** says that being in pain is occupying a certain role: caused by damage, causing distress and avoidance. What plays the role in a human could differ in an octopus. “Heart” is similar. It names the pumping role, and an artificial pump can fill it.

Frank Ramsey and later David Lewis turned this into a general method for defining a term (→ A.20, A.22).

1. **Write the job description.** Say what the thing does, using the term.
2. **Replace the term with a blank.** The sentences now describe a job without saying who does it.
3. **The term names whatever uniquely fills the blank.** If exactly one thing does the job, the term refers to it. If nothing does, the term is empty. If several things do it equally well, the term has no unique referent.

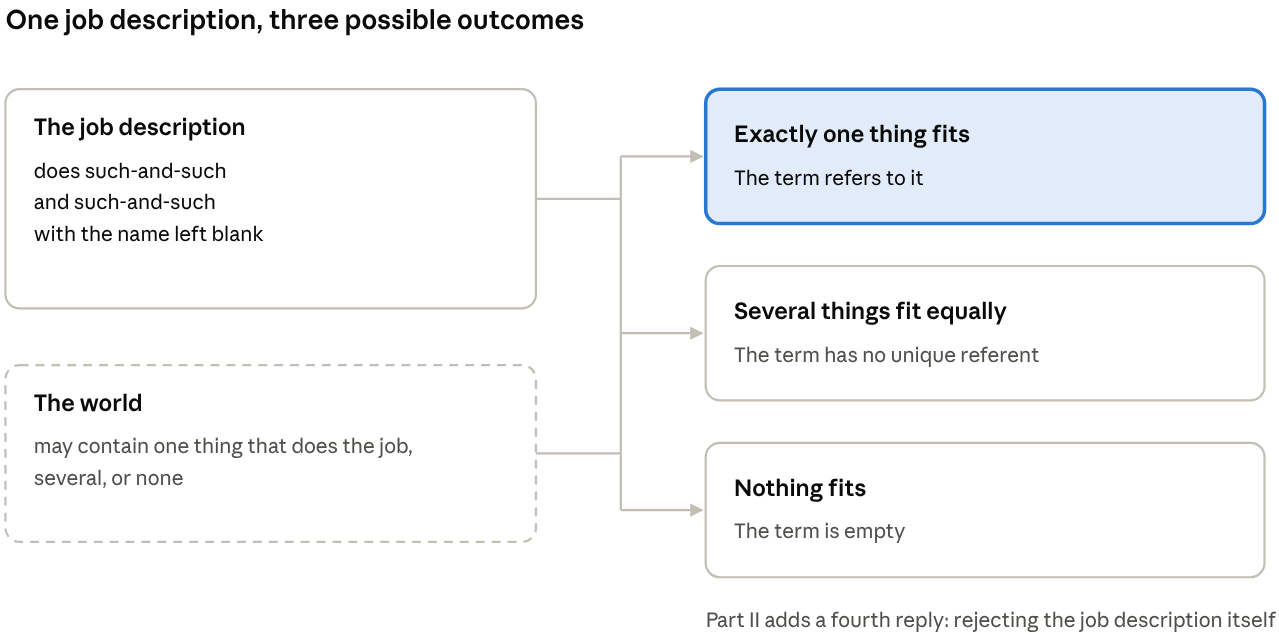


Figure 7.1 · Role and occupant, after Ramsey and Lewis

> **Ladder — the Ramsey–Lewis method**
>
> **Rung 1.** Write down everything a goalkeeper does, but leave the name blank. Whoever does all of that is the goalkeeper.
>
> **Rung 2.** Scientists did this with genes before anyone knew about DNA. "Gene" meant "whatever carries inherited traits in such-and-such a way." Later they found that stretches of DNA fill the role.
>
> **Rung 3.** Take a theory T(t) containing a new term t. Its *Ramsey sentence* is ∃x T(x): something plays the T-role. Lewis adds uniqueness: t names the one x such that T(x), and fails to refer if there is none or more than one.
>
> **Rung 4.** The "Canberra Plan" (Jackson 1998) generalizes the method: collect the platitudes about a subject, turn them into a role, and let inquiry find the occupant. Critics ask whose platitudes count. God* sidesteps that by *stipulating* a minimal role built in Chapters 2 to 6, not by collecting what people say about God. It must then show the role is one that traditions actually care about. Part III does that.

7.3 The definition

The job description is the question built in Chapters 2 to 6. Put a title on it.

> **God*** is the title of a role, not the name of a being. The role is filled by whatever, if anything, is the **non-internal ground*** of the **physical order as a whole**. That whole has two sides. **G-existence:** that there is such an order at all. **G-law:** that it is lawful. God* names **one** such ground of **both** sides.



Figure 7.2 · The definition and the keystone arch

Each phrase was built in an earlier chapter.

Phrase	Meaning	Built in
“title of a role, not the name of a being”	A job description with a blank; vacancy is a coherent answer	§7.1–7.2
“whatever, if anything”	The occupant may be a being, a principle, a value or a structure, or nothing	§3.4
“non-internal”	Not in P*, not the same as P*, and not a relocation of the question	§5.5
“ground*”	Asymmetric explanation of one of four species	§3.4
“the physical order”	P*: everything linked to paradigm physical things	§5.4
“as a whole”	The physical things, taken together	§6.2
G-existence, G-law	The two sides, each with general and specific parts	Chapter 2

Phrase	Meaning	Built in
“one … of both”	A single occupant for both sides	§7.5, Chapter 10

7.4 The formal statement

Readers new to logic can skip this section. It restates §7.3 in symbols.

Let **E** be the fact that there are physical things, and **L** the fact that they are lawful. **Int(x)** says x is internal. **G*(x, f)** says x grounds* fact f. The role is:

```
R(x) \;\equiv\; \neg \mathrm{Int}(x) \;\wedge\; G^{*}(x, E) \;\wedge\; G^{*}(x, L)
```

and the title is a definite description:

```
\mathrm{God}^{*} \;=\; \iota x\, R(x) \qquad \text{(refers only if exactly one thing satisfies } R\text{)}
```

A weaker formula picks out anything outside that grounds* *at least one* side. It is needed in Part II.

```
R^{-}(x) \;\equiv\; \neg \mathrm{Int}(x) \;\wedge\; \big(G^{*}(x, E) \,\vee\, G^{*}(x, L)\big)
```

7.5 A thin role

The role says nothing about goodness, mind, will or worship. In short, occupant theories are not the role.

In the role	Occupant theory: not in the role
Not internal to the physical order	Personal or impersonal
Grounds* that there is a physical order	Good, loving, just
Grounds* that it is lawful	All-powerful, all-knowing
One and the same for both sides	Worthy of worship
	Revealed in scripture, prophets, avatars
	Trinity, covenant, Qur’an, incarnation

In the role	Occupant theory: not in the role
	Necessary or contingent in itself
	Acts once or sustains continually

Why so thin? Thinness is what makes the vocabulary shared. In 1854 John Snow could argue with his opponents about “the cause of cholera” before anyone knew it was a bacterium. The shared term named a job. The disagreement was about what did it.

Three consequences follow. They are intended.

1. **A classical theist will say God is more than God*.** For such a view, the role being filled is necessary but far from sufficient.
2. **Someone can believe in gods while the role stands empty.** The gods of Homer and of Epicurus live inside the cosmos. They do not ground* it.
3. **Someone can reject every god while the role is filled.** A philosopher who holds that the world exists because it is good that it should has an occupant, and no deity.

7.6 The keystone

The role can be drawn as an arch. G-existence and G-law are the two feet. The physical order is the space beneath. God* is the keystone slot at the top: one stone that locks both sides, occupied or not.

One slot locks both sides, or stands empty

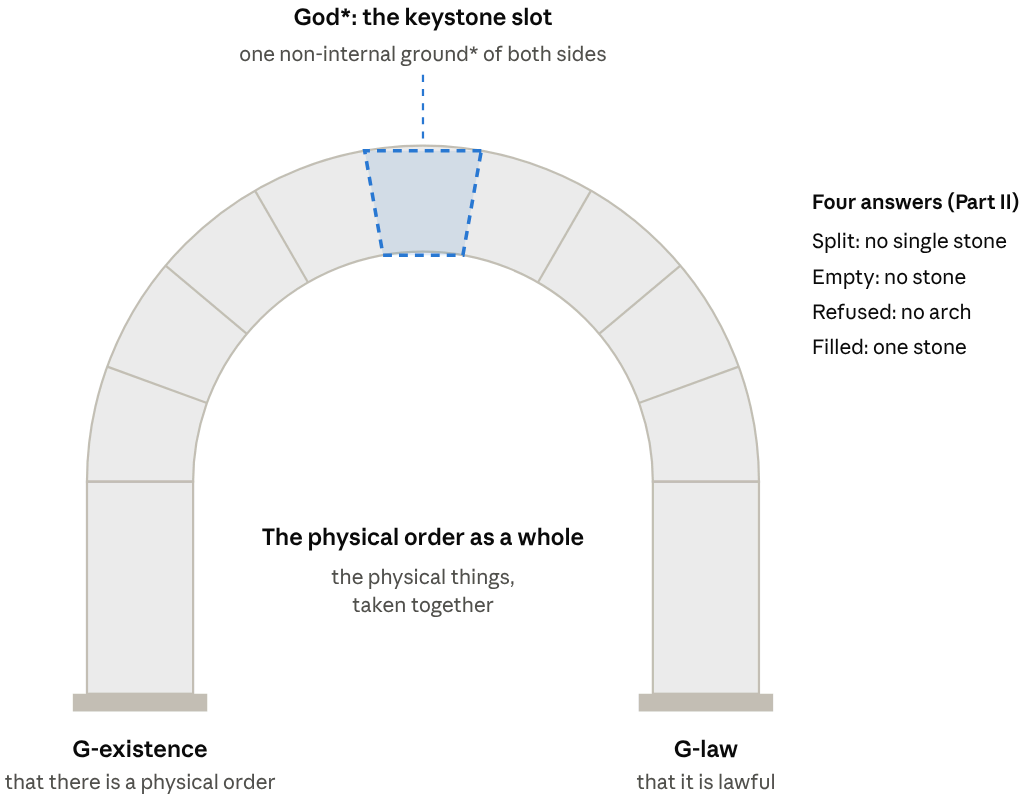


Figure 7.3 · The keystone: God* as a role over two sides

The image fits in two ways. A keystone holds the arch *now*; it is not an earlier event. In fact it is laid last, which matches Aquinas's point that dependence does not require a beginning. And some arches stand with no keystone at all. A *corbelled* arch, like the one at the Mycenaean "Treasury of Atreus," is built from stones that each overhang the one below until they meet. It is a good picture of a world that stands with nothing at the top.

Built so far: complete. *God* is the title of the role "non-internal ground* of the physical order as a whole, on both its sides."* Part I has built a question. Part II derives its possible answers.

Part II — The answers

Chapter 8 — Four verdicts, derived

The framework has four answers: Split, Empty, Refused, Filled. Their initials spell SERF, the order used throughout this book. This chapter shows they are not a list someone chose. They are the only four possibilities, and they fall out of three yes-or-no tests.

8.1 Three tests

Take any view about the world and ask three things, in order.

Test 1: Assent. Will the view affirm one of these two sentences?

- “Something non-internal grounds* at least one side of the physical order.”
- “Nothing non-internal grounds* either side.”

If it will affirm neither, because it holds that the question is ill-formed, the verdict is **Refused**. Otherwise go on.

Test 2: Outside. Does the view hold that something non-internal grounds* at least one side? Apply the internal test of §5.5 to whatever the view cites. If nothing passes, the verdict is **Empty**. Otherwise go on.

Test 3: Unique. Is there exactly one such thing, and does it ground* both sides? If yes, the verdict is **Filled**. If no, the verdict is **Split**.

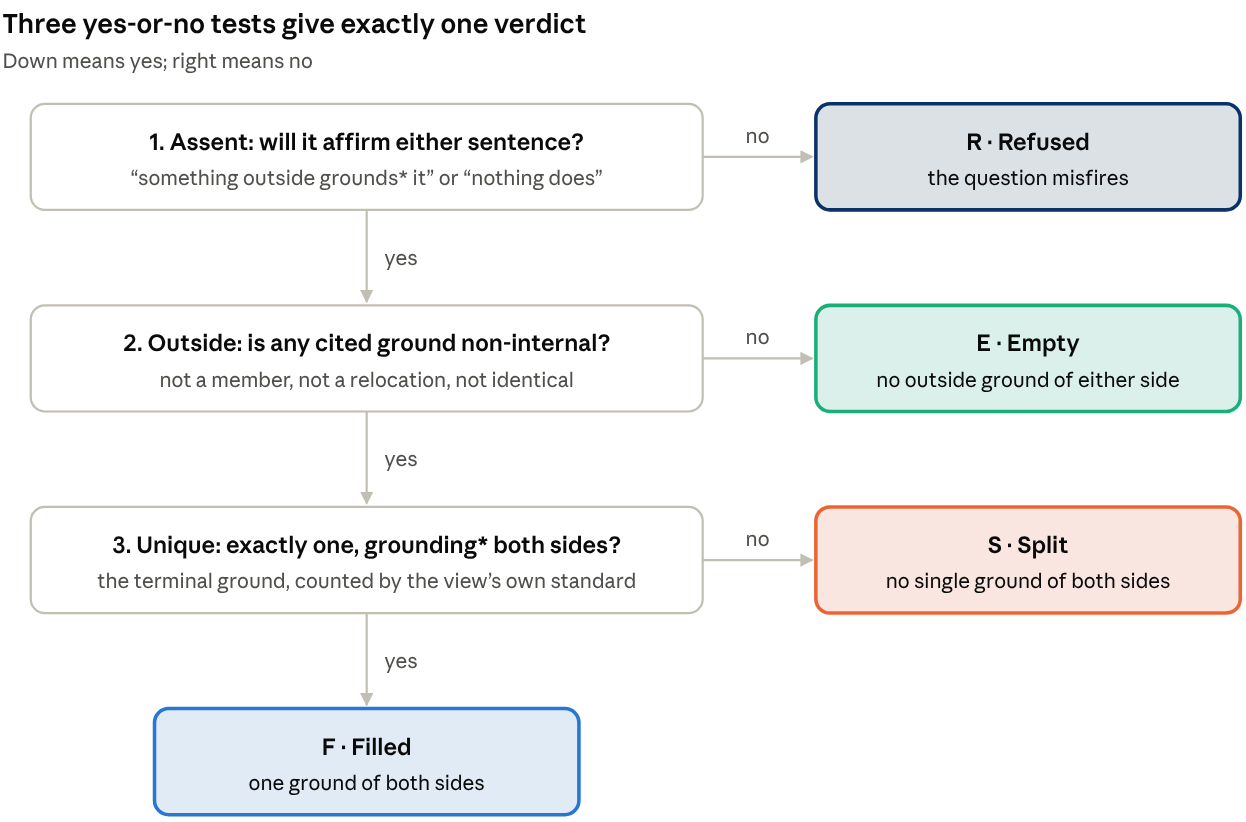


Figure 8.1 · The decision procedure

8.2 The four verdicts

Verdict	Definition	In plain words
S — Split	Something non-internal grounds* at least one side, but no one thing uniquely grounds both	Outside grounding without a single occupant
E — Empty	Nothing non-internal grounds* either side	Nothing fills the role
R — Refused	The view will affirm neither that something grounds* the order nor that nothing does	The question misfires
F — Filled	Exactly one non-internal thing grounds* both sides	One outside ground of the whole arch

The tests reach the verdicts in a different order (Refused first, Filled last). SERF is only the order for remembering and listing them.

In the symbols of §7.4: S holds when something satisfies $R^{-}(x)$ but no single thing uniquely satisfies $R(x)$. E holds when nothing satisfies $R^{-}(x)$. R holds when the view rejects the question. F holds when exactly one x satisfies $R(x)$.

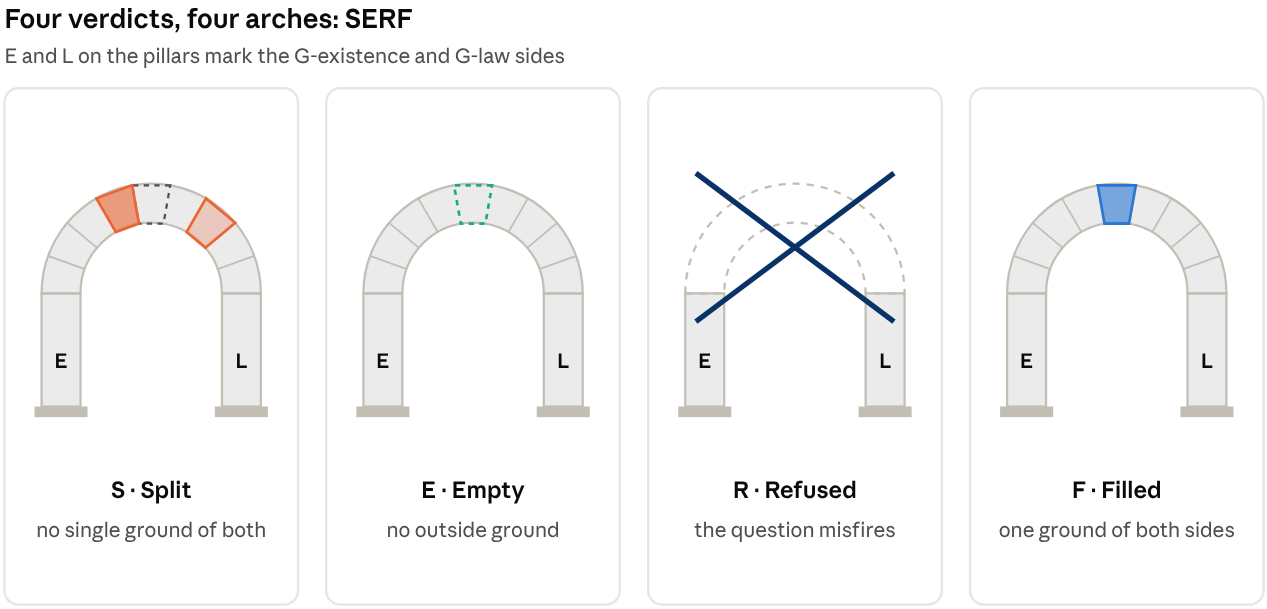


Figure 8.2 · The four verdicts as arches

8.3 Why exactly four

Claim. Any view that gives a definite answer to each of the three tests receives exactly one verdict.

Proof. Each test has two outcomes. One outcome of Test 1 is R, and the other leads to Test 2. One outcome of Test 2 is E, and the other leads to Test 3. The two outcomes of Test 3 are F and S. Every path through the tests ends at exactly one verdict, and there are no other paths. ∎

The proof is simple because the work was done in Part I. The verdicts are only as clear as the three tests, and the tests are only as clear as “non-internal,” “ground*” and “one.” That is why those terms were built so carefully.

8.4 What gets a verdict, and what does not

Three things can look like a fifth verdict. None is.

- **A view with two readings.** If a tradition's texts support two interpretations, the result is a *disjunctive* verdict, such as "Split or Filled," with each reading named. That says the view has not decided, not that there is a fifth option. Chapter 18 has many examples.
- **A person who does not know.** Not knowing is unfinished inquiry. It is not Empty, and it is not a fifth verdict. A *view* gets a verdict. A *person* can hold a **profile**: degrees of confidence across the four verdicts. An agnostic might be 10% Split, 40% Empty, 15% Refused and 35% Filled.
- **A practice that makes no claim.** A ritual, a way of life or a form of words that says nothing about the ground of the physical order is **not scored**. That differs from Refused, which is a stated position that the question misfires.

Key idea. Four verdicts for views; a profile for persons; no score for what makes no claim.

Chapter 9 — Empty and Refused

Empty and Refused both come to "no God*." They are different answers. Empty says nothing fills the role. Refused says the role misfires. The rule is to score the claim, not the tone. This chapter makes the separation exact.

9.1 The assent test at work

Put one sentence to the view: **"Nothing outside the physical order grounds* it."**

- A view that says **"True"** is Empty. It has accepted the question and answered it.
- A view that says **"That sentence is not really true or false"**, and says the same of its opposite, is Refused.

The test works because the difference between the two verdicts is a difference in what is *asserted*. "The office is vacant" is one claim. "There is no such office" is another.

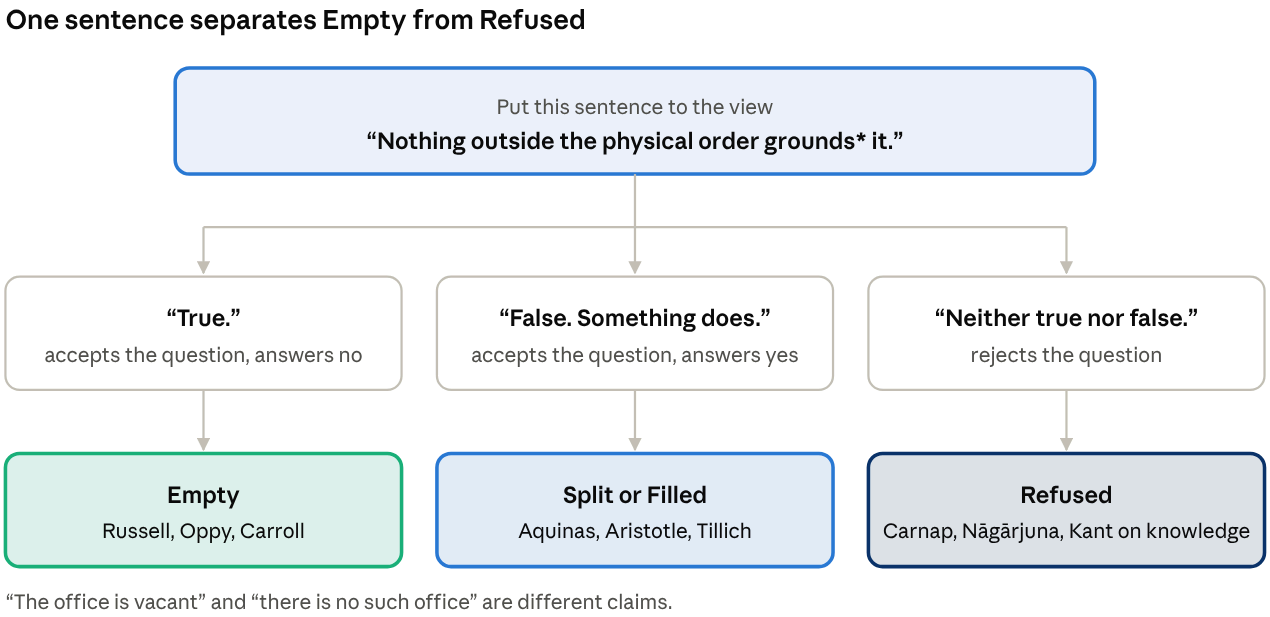


Figure 9.1 · The assent test

One consequence deserves notice. “Explanation does not apply to the universe as a whole, *so* nothing explains it” is **Empty**. The first half is the speaker’s reason. The second half is an assertion. Only a view that stops short of the assertion is Refused.

9.2 Three reasons to refuse

A refuser rejects at least one of the three presuppositions of §6.4. Each has its own tradition.

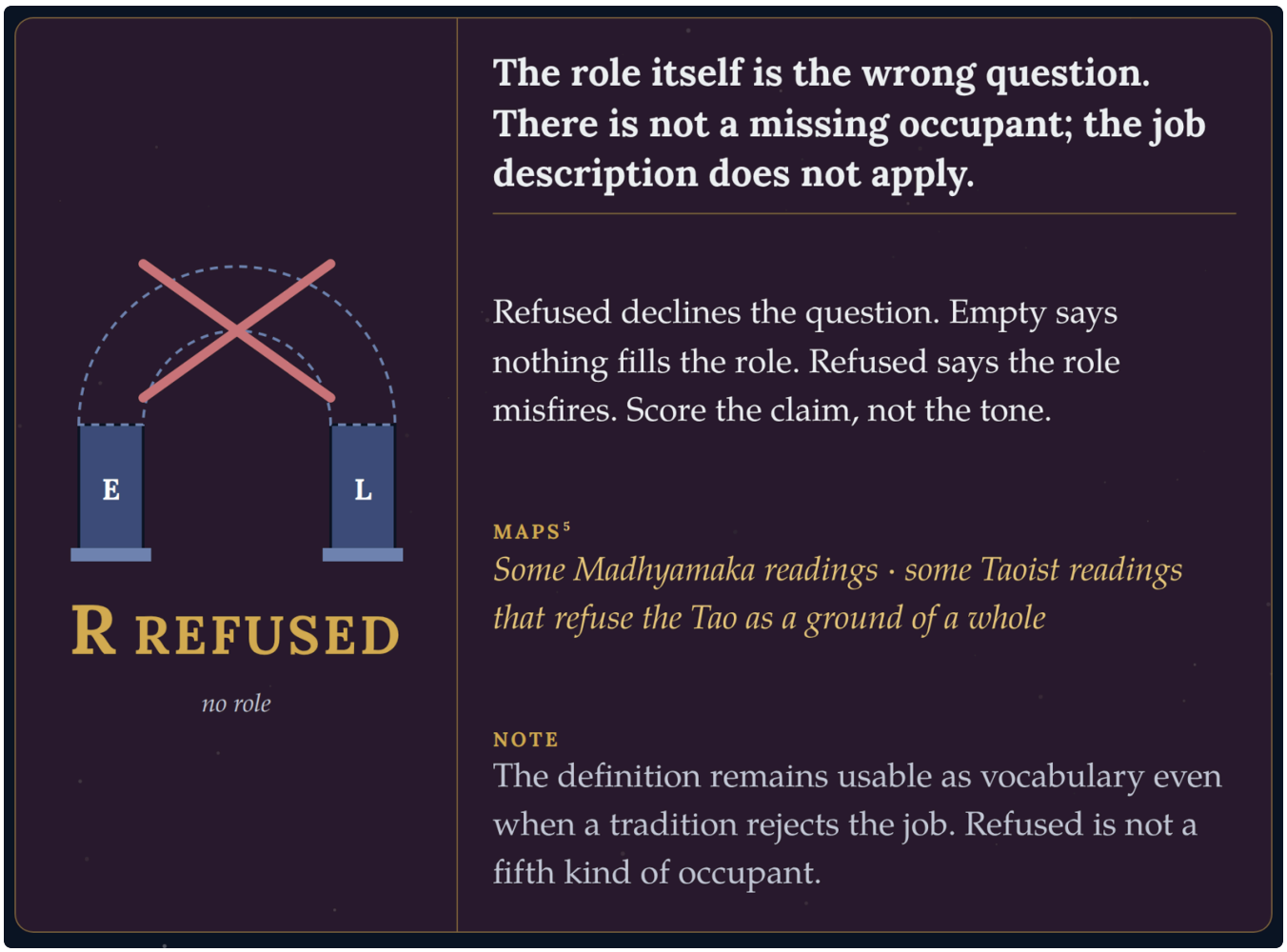


Figure 9.2 · The Refused verdict in summary

Presupposition rejected	The refusal	Who
π_1 Totality	"There is no 'world as a whole' to make claims about."	Kant on speculative reason: the world-whole is not an object of possible experience (→ A.13). Nāgārjuna: nothing, the whole included, has self-standing existence (→ A.9).
π_2 Explicability	"Questions of dependence make sense inside a framework, not about the framework."	Carnap: "external" questions are not factual questions (→ A.21). The Buddha: whether the world is eternal is among the questions set aside as undetermined (→ A.9).

Presupposition rejected	The refusal	Who
π_3 Determinacy	“Neither ‘grounded’ nor ‘ungrounded’ nor both nor neither.”	The Madhyamaka four-cornered negation. Verificationism: no possible evidence, so no content. Readings of the Dao as prior to all distinctions.

9.3 Four roads to Empty

Empty is not one position. Chapter 4 listed the ways an explanation can end inside the physical order. Each is a different Empty.

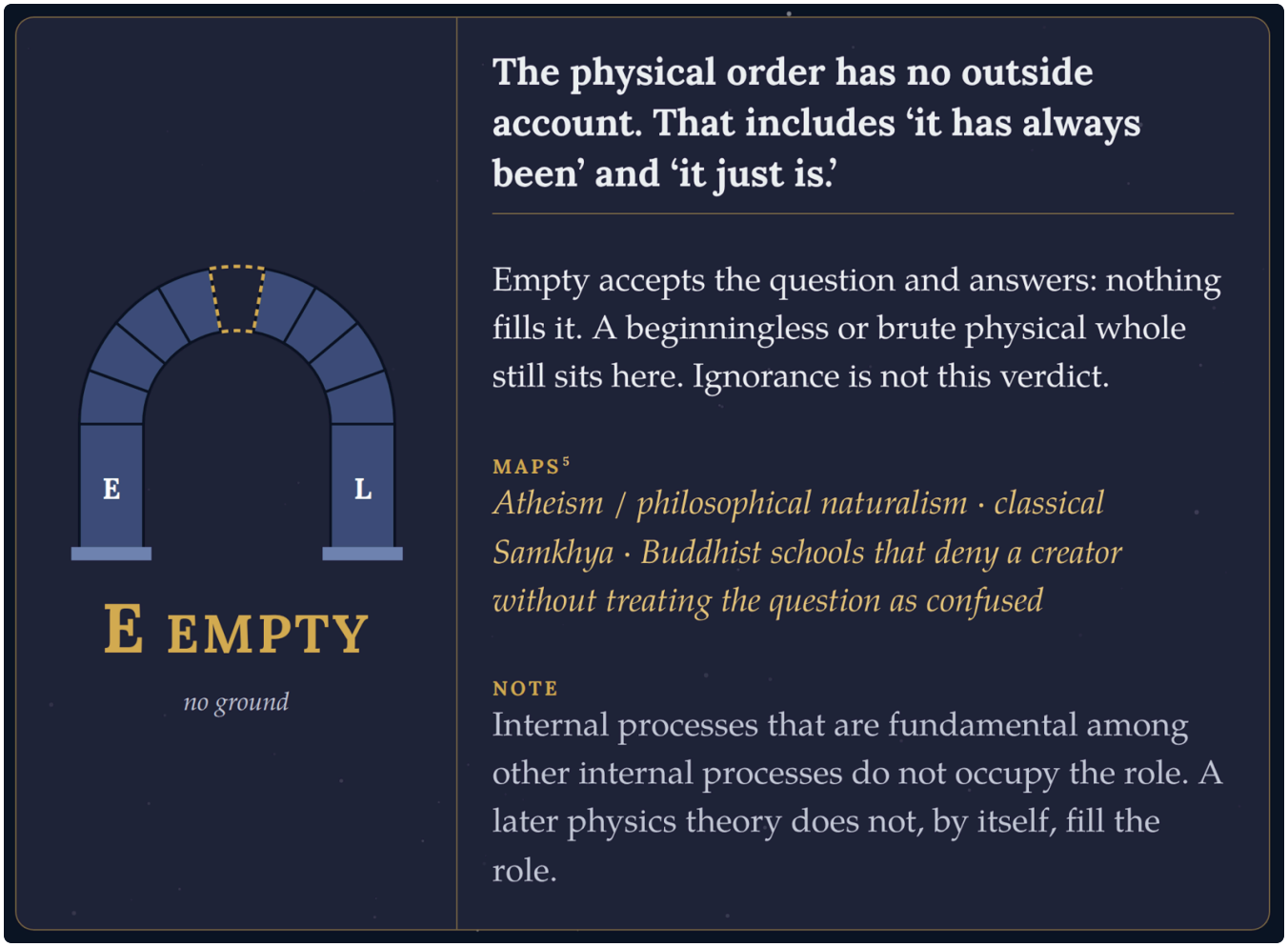


Figure 9.3 · The Empty verdict in summary. The table below divides it into four kinds.

Kind of Empty	Claim	Examples
Empty (brute)	The order exists and is lawful for no reason	Russell's "just there"; Sean Carroll's "the universe simply is"
Empty (necessary)	The order, or its first state, could not have failed to exist	Oppy's necessary initial state
Empty (regress)	Each state is explained by an earlier one, without end, and nothing more is needed	Past-eternal cosmologies taken as complete explanations
Empty (identity)	What is offered as the ground is the order itself, or part of it	Stoicism; priority monism; Tegmark's mathematical universe

The rule that necessity given the laws does not fill the role generalizes. No grade of necessity of the physical order fills the role, because necessity is a status of the order, not a thing outside it. What necessity can do is make the role idle. That is still Empty.

9.4 A worked example: Russell in 1948

In the BBC debate, Russell said things of both kinds, which is why readers disagree about where he stood.

What Russell said, in paraphrase	What it is
"I should say that the universe is just there, and that's all."	An assertion: nothing grounds it
The concept of cause does not apply to the total	A reason offered for that assertion
The human race has no mother	An argument that the inference from parts to whole fails
"Necessary" applies to propositions, not to beings	An objection to one occupant theory

Apply the assent test. Russell affirms that the universe is "just there." He accepts the sentence "nothing outside grounds it." His verdict is **Empty (brute)**. His sceptical remarks about cause are his reasons, not a withdrawal.

Compare Carnap, who would decline to call either sentence true, and Nāgārjuna, who explicitly rejects all four corners. They are Refused.

9.5 Three look-alikes

1. **Unknowing.** The *Nāsadīya* hymn's "perhaps he does not know" is humility about the answer. It is a profile, not a verdict.
2. **Kant.** As a matter of *knowledge*, Kant refuses: speculative reason cannot reach the world-whole. As a matter of *practical* reason, he postulates God. These are two claims and they get two scores: Refused, and Filled. Thinkers are not always one view.
3. **Negative theology.** Pseudo-Dionysius and Tillich deny that God is *a* being, or "exists" in the ordinary sense (→ A.14). That can sound like refusal. But both affirm a source of all that is. They pass the assent test on the "something" side. Their verdict is Filled.

9.6 A phrasebook

If a view says…	Verdict
“Something beyond orders it; its existence is eternal or brute.”	Split (law-side)
“Something beyond gives it existence; its order has no outside source.”	Split (existence-side)
“Two things beyond it, one for each.”	Split (two grounds)
“Nothing beyond it. It just is.”	Empty (brute)
“Nothing beyond it. It had to be.”	Empty (necessary)
“What you call its ground is the world itself.”	Empty (identity)
“There is no whole to ask about.”	Refused (π_1)
“That is not a factual question.”	Refused (π_2)
“Not grounded, not ungrounded, not both, not neither.”	Refused (π_3)
“One thing beyond the physical makes it exist and makes it orderly.”	Filled
“We cannot know.”	A profile
Nothing on the subject	Not scored

Chapter 10 — Split and Filled

10.1 What Filled requires

Filled needs one thing that is non-internal and grounds* both sides. It does not need that thing to be a person, a mind, good, or an object of worship. Filled can be personal or not.

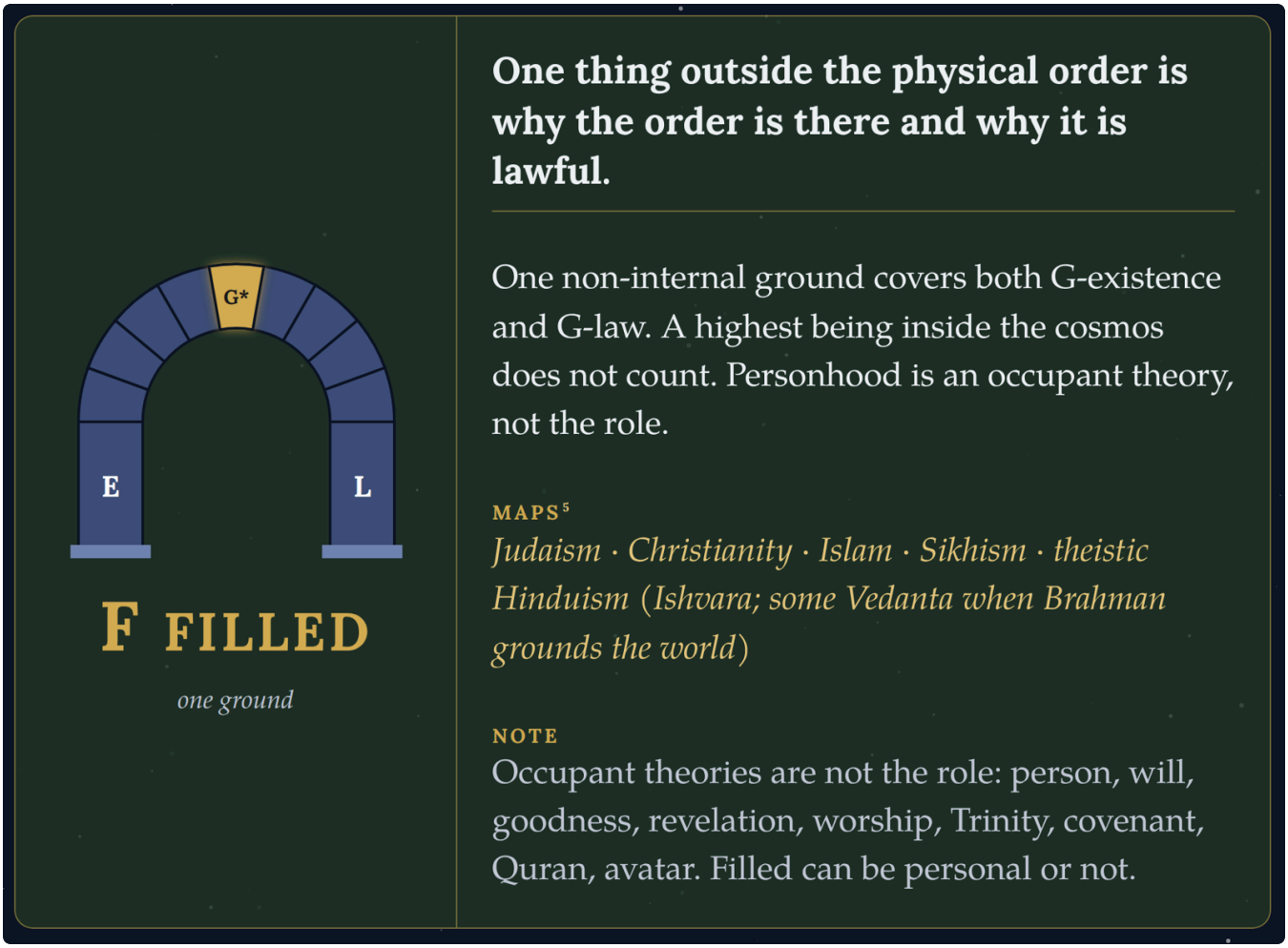


Figure 10.1 · The Filled verdict in summary

Three examples show the range.

- **Classical Christian theism:** a personal creator who produces the world from nothing, legislates its laws, and sustains it at every moment.
- **Plotinus:** an impersonal One from which everything flows without choice (→ A.5).
- **Axiarchism:** no being at all, only the ethical requirement that a good world exist (→ A.23).

All three are Filled. They differ in the occupant theory.

10.2 Chains: the terminal clause

Many systems place several non-internal items in a chain. Plotinus has the One, then Intellect, then Soul, then the world. Some theologians say God's ideas ground the laws and God grounds the ideas. Ground* carries along a chain, so every link would count as a ground* of the order, and the role would look crowded.

The terminal clause. The occupant is the **terminal** non-internal ground*: the one not itself grounded* by a further non-internal item. The links between it and the physical order are *channels*. They are recorded, but they are not rivals.

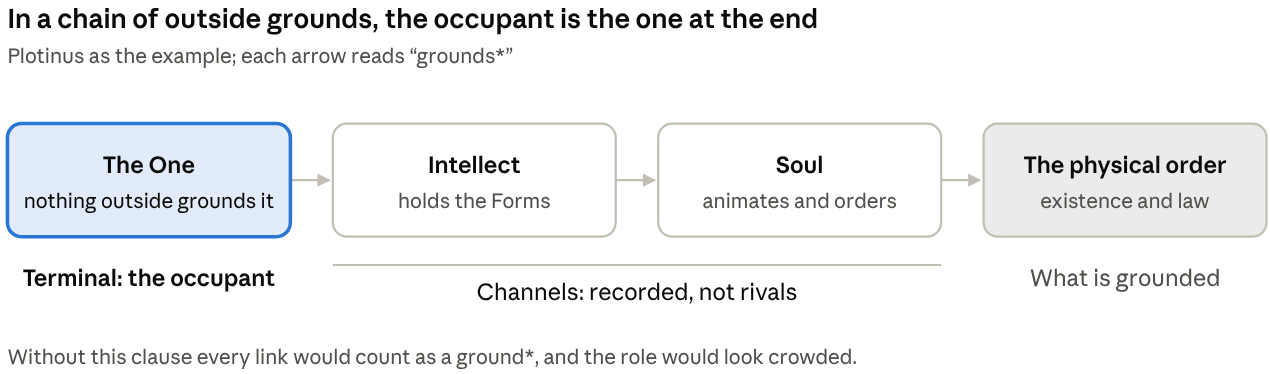


Figure 10.2 · The terminal clause

This also handles an old puzzle. If a view says that goodness makes God what God is, and God makes the world, then the Good is terminal and God is its channel. If the view says God and God’s goodness are identical, there is one occupant.

10.3 Counting: when is there “one”?

Count grounds by the view’s *own* standard of identity. Classical Christian theology holds that the three persons of the Trinity act toward creation as one (“the external works of the Trinity are undivided”). That is one ground, and Filled. A pantheon in which different gods each made a different part of the world is several.

10.4 What Split is

Split means: outside grounding without a unique occupant of both sides. It has four main forms.

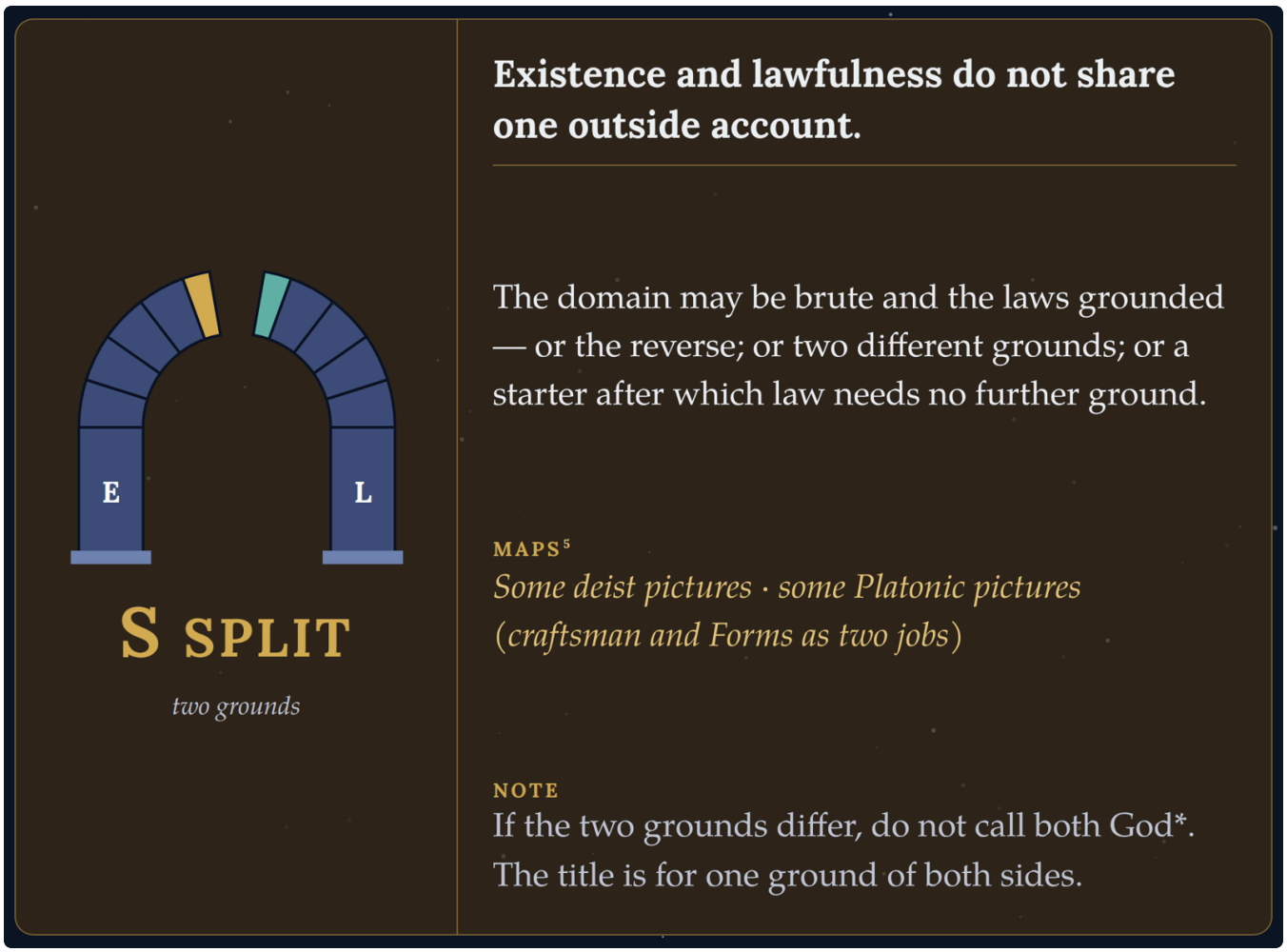


Figure 10.3 · The Split verdict in summary. Section 10.5 revises the placement of deism shown here.

Form	Situation	Example
Split (law-side)	Something outside grounds* the order's lawfulness; its existence is brute, eternal or internal	Aristotle's unmoved mover; Nyāya's Lord arranging eternal atoms; process theism
Split (existence-side)	Something outside grounds* that there is stuff; its regularity has no outside source	A maker of matter whose laws are primitive and unmade
Split (two grounds)	Different outside things ground* each side, or share one side	Plato's craftsman and Forms
Split (joint)	Several outside things ground* the order together, none alone	Manichaeism's Light and Darkness

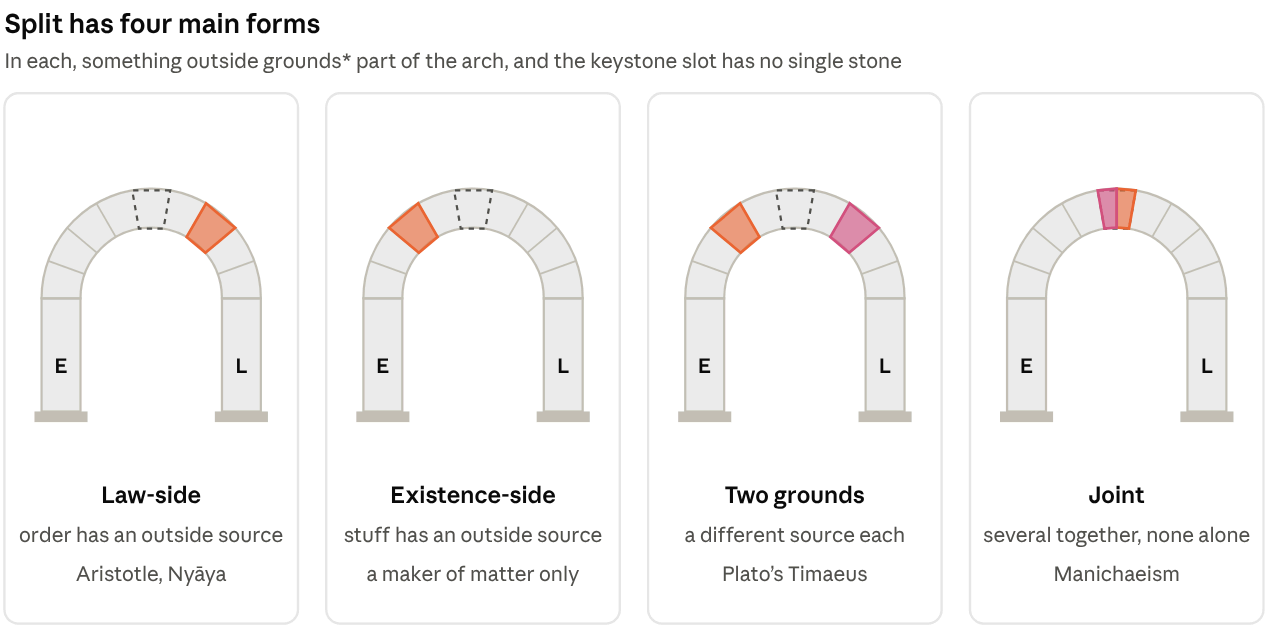


Figure 10.4 · The four main forms of Split

Two rare boundary cases are placed in Split by rule and flagged as such: a role with *two or more complete, independent occupants*, and a chain of non-internal grounds that *never terminates*. In both there is outside grounding and no unique occupant, which is what Split means, though neither is a split in the everyday sense of the word.

Which forms of Split are available depends on the theory of laws (§2.4). On a powers view, whoever grounds* the existence of the propertied things thereby grounds* the laws, so "Split (existence-side)" is closed. "Split (law-side)" stays open on every theory, because an orderer can always work on stuff it did not make.

10.5 Once or always

An earlier version of the framework listed "a starter after which law needs no further ground" under Split. On the definitions built here, that is not quite right. The point is worth care, because the constraint that the ground is not an earlier event in time might seem to support it.

CONSTRAINTS

Ground is not an earlier event in time, not a first particle, and not a piece of the order. Being fundamental among physical processes does not fill the role. Necessity given the laws does not fill the role. Personhood and worship-worthiness are claims about an occupant, not part of the role. Not knowing is unfinished inquiry; it is not Empty and not a fifth verdict.

Figure 10.5 · The constraints on the role

The constraint excludes an early *physical event*, such as a first state, from being the ground. It does not exclude a non-internal *agent* whose act happens to be a single one. A deist God who brings into being both matter and its laws, and then lets them run, is one outside ground of both sides. That is Filled. Whether the ground acts once or sustains continually is part of the occupant theory.

A starter is Split only if it grounds* one side and not the other: if it makes the stuff but the laws are independent of it, or orders stuff it did not make.

Chapter 11 — Core and extensions

A framework that grows a label for every case becomes impossible to test. This chapter draws a line. The **core** is small and fixed. The **extensions** are optional and exist to record detail.

11.1 The core

Core element	Content
One role	God*: the non-internal ground* of the physical order as a whole
Two sides	G-existence and G-law
Three tests	Assent; Outside; Unique
Four verdicts	Split, Empty, Refused, Filled (SERF)

Everything a reader needs to use the framework is in that table.

11.2 The extensions

Extension	What it records	Example
Forms of Split	Which side, or how many grounds	Split (law-side)
Kinds of Empty	Which ending	Empty (necessary)

Extension	What it records	Example
Species tags	The kind of ground*	productive, constitutive, nomic, axiological
Occupant tags	Personal or impersonal; concrete or abstract	impersonal
Mode tags	Acts once, or sustains	sustaining
“Generic only”	The ground covers that there is law, not which laws	—
Levels rule	For traditions with levels of truth: score at the level where the tradition places the physical order	Advaita, §14.5
Profiles	A person’s confidence across the verdicts	10 / 40 / 15 / 35
Wider roles	A larger domain to be grounded	below

A full entry reads like this: *Aquinas: Filled [productive, nomic; personal; sustaining].*

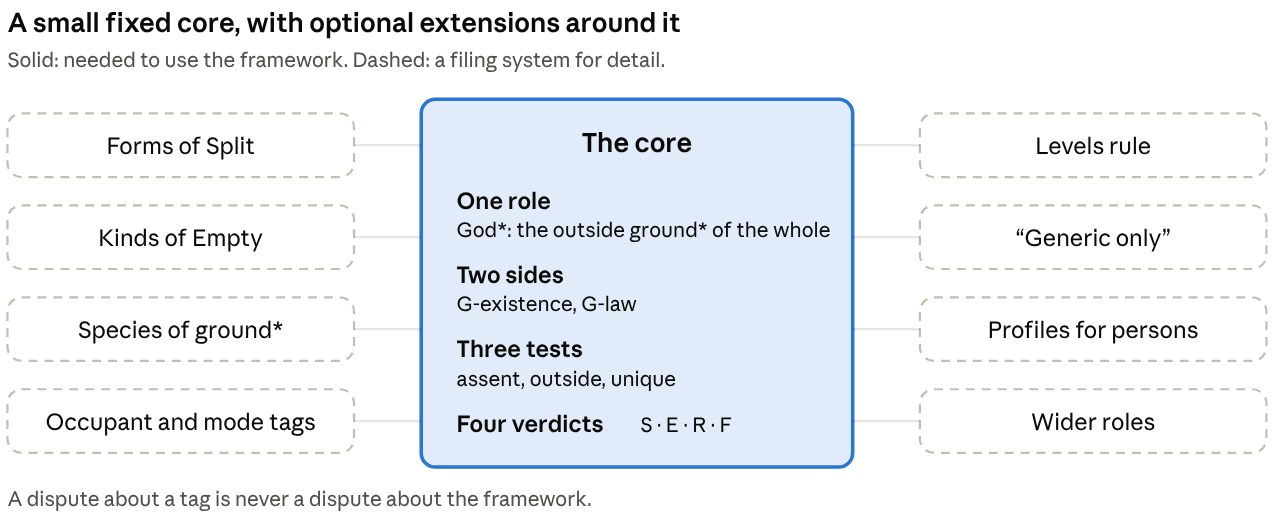


Figure 11.1 · Core and extensions

11.3 Wider roles

Some inquiries need a larger domain than the physical order.

- **God* (concrete):** the non-internal ground* of all concrete reality, physical and mental.

- **God* (all):** the non-internal ground* of concrete reality and of abstract objects such as numbers.

Why not use the widest role from the start? Because nothing is outside *everything*. If the domain were "all that exists," the occupant would be inside it by definition, and the role could never be filled. Theists do not say God is outside reality. They say God is outside creation. A useful role needs a domain whose outside is an open question, and the physical order is the one domain every party grants.

The wider roles show where views part. A theist who holds that numbers exist independently of God is Filled for the basic role and Split for the widest one.

11.4 The scoring protocol

1. **State the claim.** Quote or closely paraphrase what the view says about why there is a physical order and why it is lawful. Cite a primary text or a recognized commentator. No claim: not scored.
2. **Fix the level,** if the tradition distinguishes levels of truth.
3. **Apply the assent test.** If the view affirms neither sentence, the verdict is Refused. Note which presupposition it rejects.
4. **Apply the outside test.** List what the view offers as ground. Test each item with (a), (b) and (c) of §5.5, using the intrinsic-description rule.
5. **Apply the unique test.** Find the terminal ground. Ask whether it grounds* both sides. Count by the view's own standard.
6. **Record** the verdict, any extension tags, any rival reading, and what evidence would overturn the entry.

Key idea. The core fits on a card. The extensions are a filing system. A dispute about a tag is never a dispute about the framework.

Part III — The question in history

Part I built the question by reasoning alone. Part III checks it against the record. If the distinctions are real, the history of thought should show people drawing them, and it does. Thinkers in every major civilization asked why there is a world and why it is orderly, and many answered one question and not the other. Each section below ends with the verdict the framework gives. Appendix A has a primer on each figure, and Chapter 18 collects the verdicts.

Figure 12.1 · Three thousand years of the question

Chapter 12 — The ancient and medieval worlds

12.1 Greece: stuff and pattern

The first Greek philosophers, in the sixth century BCE, asked for the *archē*, the origin or principle of all things (→ A.1).

- **Thales** said water.
- **Anaximander** said the *apeiron*, the boundless.
- **Heraclitus** said the world is ever-changing fire, whose changes follow a *logos*, a rational measure.

Two kinds of answer are already present: a *stuff* the world is made of, and a *pattern* it follows. They are the ancestors of the two sides.

Parmenides added the principle later summed up as *ex nihilo nihil fit*: nothing comes from nothing. Almost every later thinker had to respond to it.

The atomists (Leucippus, Democritus, later Epicurus and Lucretius) gave the first fully internal account: eternal atoms moving in void, with no designer. Their gods exist, but inside the cosmos and made of atoms. *Verdict: Empty (brute), with gods.*

12.2 Plato

Plato's *Timaeus* tells a "likely story" (→ A.3). A divine craftsman, the **Demiurge**, looks to the eternal **Forms** as a model and brings order to a chaotic, pre-existing "receptacle." He does not make the raw material, and he does not make the Forms. In the *Republic*, the Form of the Good is "beyond being" and is the source of the other Forms' being and intelligibility.

Read through the *Timaeus*, Plato has two outside grounds of order and none of the receptacle's existence. *Verdict: Split (two grounds).* Read through the *Republic*, with the Good as the one source of everything: *Filled [axiological, nomic]*. The framework does not choose between the readings. It shows what the choice is about.

12.3 Aristotle

Aristotle held that the cosmos has always existed (→ A.4). Motion still needs explaining, so he posited an **unmoved mover**: pure actuality, "thought thinking itself." It moves the heavens as an object of love moves a lover, as a final cause. It does not create matter or

explain why there is a cosmos at all. *Verdict: Split (law-side) [axiological; impersonal].* Aristotle is the clearest ancient case of a ground of order with no ground of existence.

12.4 Stoics and Plotinus

The Stoics identified God with the *logos*, a fiery, rational, *bodily* principle pervading the cosmos (→ A.5). God is the world's mind, made of the world's stuff. By test (a) of §5.5 that is internal. *Verdict: Empty (identity).* The result is instructive: a devoutly theistic school lands in Empty, because its God is not outside the order.

Plotinus went the other way. Everything flows from **the One**, a principle so simple it is beyond being. From the One comes Intellect, then Soul, then the physical world. *Verdict: Filled [productive, by emanation; impersonal].* By the terminal clause, the One is the occupant, and Intellect and Soul are channels.

12.5 The Abrahamic traditions

The Hebrew Bible opens with God creating "the heavens and the earth." Scholars debate whether Genesis 1 describes creation from nothing or the ordering of a watery chaos. The second reading would be Split (law-side); the first, Filled. A verse in 2 Maccabees (7:28) is often cited as an early statement of creation from nothing, though Gerhard May (1994) argues that the doctrine was only clearly formulated by Christian writers of the later second century, such as Theophilus of Antioch and Irenaeus, in argument with Platonist views of eternal matter. From then on, Judaism, Christianity and Islam taught that one God is the source of both the world's existence and its order.

Four ideas from this tradition are built into the definition.

- **Augustine** argued that God made time along with the world (→ A.6). There was no "before." A ground is not an earlier event.
- **Avicenna** distinguished a thing's *essence* (what it is) from its *existence* (that it is) (→ A.7). Anything whose essence does not include existence needs a cause of its existing. The chain ends in the **Necessary Existent**. This is the clearest source for the existence side.
- **Al-Ghazālī** argued that the regularities of nature are God's habits, not necessary connections. That makes God the direct ground of the law side.
- **Aquinas** argued that reason cannot prove the world had a beginning, and that a beginningless world would *still* depend on God at every moment (→ A.8).

Verdict for the classical forms of all three traditions: Filled [productive, nomic; personal; sustaining]. The Sikh *Mūl Mantar*, which opens the Guru Granth Sahib with one timeless, self-existent creator, fits the same pattern (→ A.9).

12.6 India

Indian philosophy holds the widest range of positions of any single civilization (→ A.9).

- **The Vedas** speak of *ṛta*, the cosmic order: an early concept of the law side. The *Nāsadīya* hymn asks how creation arose and ends in open doubt. *Not a verdict: a profile.*
- **The Upaniṣads** name **Brahman** as that from which beings arise, by which they live, and into which they return.
- **Classical Sāṃkhya** explains the world as the unfolding of primal matter (*prakṛti*) in the presence of pure consciousness (*puruṣa*), with no creator. *Empty (brute).*
- **Nyāya-Vaiśeṣika.** Udāyana, around 1000 CE, argued for a Lord, **Īśvara**, who arranges atoms that are themselves eternal and unmade. *Split (law-side) [productive; personal].*
- **Mīmāṃsā and Jainism** argue against a creator. The universe is beginningless and runs by its own laws. *Empty.*
- **Early Buddhism.** The Buddha set aside, as "undetermined," questions such as whether the world is eternal. *Refused (π_2).* Later Buddhist philosophers such as Dharmakīrti argued directly against a creator, which is to accept the question and answer it. *Empty.*
- **Madhyamaka.** Nāgārjuna argued that nothing has *svabhāva*, self-standing existence. There is no self-standing whole to be grounded and no self-standing ground. *Refused (π_1, π_3).* Chapter 15 studies him closely.
- **Vedānta** divided over how Brahman relates to the world. For Śaṅkara's *Advaita*, Brahman alone is ultimately real and the world is its appearance. For Rāmānuja, the world is Brahman's body. For Madhva, Brahman is a distinct creator. *All three: Filled, on the reading defended in §14.5; Advaita's placement is disputed.*

12.7 China

The *Daodejing* opens by saying that the Dao that can be spoken is not the constant Dao (→ A.9). Chapter 42 says "the Dao gives birth to one, one to two, two to three, three to the ten thousand things." Read as an account of origins, the Dao grounds both the existence and the patterning of things: *Filled [productive, constitutive; impersonal]*. Read as the way

things themselves go, prior to every distinction including “grounded” and “ungrounded”: *Refused (π_3)*. The commentator **Guo Xiang** (d. 312) took a third line: things are “self-so” and nothing produces them. *Empty (brute).*

Zhu Xi (1130–1200) gave the closest historical parallel to the two sides. Everything is *qi*, the material force that makes it concrete, and *li*, the principle or pattern that makes it what it is. All li are one in the **Supreme Ultimate**. Qi corresponds to G-existence and li to G-law. Whether the Supreme Ultimate grounds qi as well as li is disputed. *Split (law-side) or Filled.* Chapter 15 examines the texts.

The historian Joseph Needham (1951) argued that Chinese thought pictured natural order as an organic harmony, not a code issued by a lawgiver. “Law of nature” is itself a metaphor with a history.

Chapter 13 — From the scientific revolution to today

13.1 Laws, clocks and sufficient reason

The scientific revolution made laws of nature central, and with them the law side (→ A.10, A.11).

- **Descartes** derived the laws of motion from God’s unchanging nature.
- **Boyle and Newton** spoke of laws as God’s decrees. Newton inferred an intelligent lawgiver from the order of the solar system.
- **The Leibniz–Clarke correspondence** (1715–16) turned on *how* God grounds law. Leibniz mocked a God who must rewind his clock.
- **Deism** pictured a creator who makes the world and its laws and lets them run. *Filled [productive, nomic; personal; once]* (§10.5).
- **Spinoza** identified God with Nature: one substance of which everything is a mode. *Filled on the standard reading; Empty (identity) on a naturalist one* (§14.2).
- **Leibniz** stated the principle of sufficient reason and asked why there is something rather than nothing. *Filled [productive, nomic, axiological; personal]*: God chooses the best possible world.

13.2 The critics

Hume’s *Dialogues* (1779) attacked both the design argument and the argument from contingency (→ A.12). His character Philo suggests that the material world may contain the principle of its own order. Hume’s own final position is debated. *Empty, or a sceptical profile.*

Kant (1781) argued that the cosmological argument secretly depends on the ontological one, and that reason falls into contradictions when it asks whether the world has a beginning (→ A.13). *Refused, for knowledge; Filled, as a practical postulate.*

In the nineteenth century, Laplace is said to have told Napoleon he had "no need of that hypothesis." Darwin removed the need for a designer of organisms, though not of laws. **Hegel** described reality as Spirit unfolding itself; his followers split over whether that is theism or naturalism. *Filled [constitutive] or Empty (identity), by reading.*

13.3 The twentieth century

- **The logical positivists** declared the question meaningless (→ A.21). *Refused.*
- **Paul Tillich** said God is not a being but *being-itself*, the "ground of being" (→ A.14). *Filled [constitutive].*
- **Martin Heidegger** made "Why are there beings at all instead of nothing?" the first question of metaphysics.
- **Alfred North Whitehead** made God one factor among others: God supplies the world's order and aims, while its sheer ongoing creativity is not God's doing (→ A.15). *Split (law-side).*
- **Bertrand Russell** said the universe is "just there." *Empty (brute)* (§9.4).

13.4 The contemporary scene

From the 1960s, analytic philosophers such as Alvin Plantinga, Richard Swinburne and William Lane Craig rebuilt the classical arguments with modern logic (→ A.16). Naturalists such as Graham Oppy replied with equal care. Three further developments matter.

1. **Grounding theory** supplied a vocabulary for non-causal dependence (Chapter 3).
2. **Physical cosmology** supplied real data about the early universe (Chapter 5).
3. **Global philosophy of religion** began testing its categories on non-Western traditions (Chapter 19).

13.5 What the history shows

Thinker or school	Outside ground of existence?	Outside ground of order?	Verdict
Atomists	None	None	Empty
Plato (*Timaeus*)	None	Craftsman and Forms	Split

Thinker or school	Outside ground of existence?	Outside ground of order?	Verdict
Aristotle	None	Unmoved mover	Split
Stoics	None: the divine is inside	None: the logos is inside	Empty
Plotinus	The One	The One	Filled
Avicenna, Aquinas	God	God	Filled
Nyāya	None: atoms are eternal	Īśvara	Split
Zhu Xi	Disputed	The Supreme Ultimate	Split or Filled
Whitehead	None: creativity is ultimate	God	Split
Nāgārjuna	Question rejected	Question rejected	Refused

Key idea. The two sides have separate histories. Great systems in Greece, India, China and the modern West ground the order of the world and not its existence. A vocabulary that cannot say so is too blunt, and the Split verdict is how this one says it.

Part IV — Testing the framework

Chapter 14 — Hard cases

A definition is tested at its edges. Each case below is one that could break the framework. For each: the case, why it is hard, how the three tests resolve it, and the verdict.

14.1 The simulation hypothesis

Case. Nick Bostrom (2003) argued that we may be living in a computer simulation run by an advanced civilization. The simulator would be responsible for both the existence and the laws of our universe.

Why it is hard. A programmer filling the role that the One and the Necessary Existent were proposed for would be absurd. A framework that allowed it would be measuring the wrong thing.

Resolution. The simulator's computer is internal twice over. By the realization link of §5.4, the computer that our world runs on belongs to P*, and so does everything that interacts lawfully with that computer, including its builders. And by the relocation test, a computer in a lawful world is itself a regular domain of states, so citing it only moves the two-sided question down a floor.

Verdict. The simulation hypothesis is **verdict-neutral**. It changes the size of P*, not what grounds it. All four verdicts remain open about the larger order. David Chalmers (2022) reaches a compatible conclusion from another direction: a simulated world is a real world, and its simulator is at most a "local" creator.

14.2 Spinoza, and the Stoics beside him

Case. For Spinoza there is one substance, "God or Nature," with infinitely many attributes, of which we know two: extension (the physical) and thought. Everything is a *mode* of it. God is "the immanent, not the transitive, cause of all things" (*Ethics* I, proposition 18).

Why it is hard. The Stoics also identified God with nature, and they scored Empty. Why should Spinoza differ?

Resolution. Apply the three clauses of the internal test, using the intrinsic-description rule.

	Stoic logos	Spinoza's substance
What the view says it is	A fine, fiery body spread through the cosmos	A self-existent substance with infinitely many attributes; the physical is one of them
(a) Member of P*?	Yes: a body among bodies	No: not a body, and not at any place
(b) Relocates the question?	—	No: by Spinoza's account it exists by its own nature and is the source of all laws, not subject to them
(c) Identical to P*?	—	No: P* is the substance under *one* attribute, in the form of its modes
Result	Internal	Non-internal

"Non-internal" does not mean "located elsewhere." It means not a member, not a relocation, and not the same thing.

Verdict. Spinoza, standard reading: **Filled [constitutive, nomic; impersonal]**. A *naturalist* reading, which drops the further attributes and treats "God" as an honorific for physical nature, gives **Empty (identity)**. The Stoics: **Empty (identity)**.

14.3 Priority monism

Case. Jonathan Schaffer (2010) argues that the cosmos as a whole is the one fundamental thing, and its parts depend on it.

Resolution. That is a claim about dependence *within* P*: the whole over its parts. The whole is P* itself, and nothing is its own outside ground.

Verdict. Empty (identity). The lesson: being fundamental among physical processes does not fill the role.

14.4 A necessary cosmos

Case. Graham Oppy suggests that the first state of the natural world might be necessary. Then nothing outside is needed, and even the principle of sufficient reason is satisfied.

Why it is hard. "It had to be" is a kind of explanation. Does it count as an account from outside?

Resolution. Necessity is a status the order has, not a thing beyond it. Nothing non-internal is cited. The view passes the assent test on the "nothing" side.

Verdict. Empty (necessary). Chapter 15 returns to Oppy.

14.5 Advaita Vedānta

Case. For Śaṅkara, Brahman alone is ultimately real. The world is real for everyday purposes but is an appearance superimposed on Brahman. The *Brahma Sūtras* define Brahman as "that from which the origin, and so on, of this [world] proceed" (1.1.2).

Why it is hard. At the ultimate level there is no world to ground, which sounds like rejecting totality (π_1), and so Refused. At the everyday level Brahman is plainly the world's source, which sounds like Filled.

Resolution. Use the levels rule of §11.2: score a tradition at the level where it places the physical order. Advaita places the physical order at the everyday level, and at that level it *affirms* a ground: Brahman is the world's material and efficient cause. It passes the assent test on the "something" side. That the world is ultimately not other than Brahman is a claim about what the grounded thing finally is. It belongs to the occupant theory.

Verdict. Filled [constitutive, productive], marked *disputed*. A scholar who holds that Śaṅkara's final teaching cancels the everyday-level claim would score Refused. Rāmānuja and Madhva are Filled without dispute.

Contrast with Madhyamaka. Nāgārjuna also distinguishes two levels. But at *neither* level does he affirm a ground, or affirm that there is none. That is the difference, and the assent test detects it.

14.6 Axiarchism: a value in the keystone

Case. John Leslie (1979) and Nicholas Rescher (2010) propose that the world exists *because it is good that it should*. The ethical need for a good universe is itself creative. No person, mind or substance is involved.

Why it is hard. Can an abstract requirement ground* anything?

Resolution. Whether it *can* is a question about whether the answer is true. The framework asks only whether the view claims a non-internal ground* of both sides. It does, of the axiological species. "Whatever, if anything" in the definition was written for this.

Verdict. Filled [axiological; abstract; impersonal]. It is the clearest case of a filled role with no god.

14.7 Idealisms

Idealists hold that the physical depends on mind. The verdict turns on whether that mind is outside the physical order or is the order's own inner nature.

View	Claim	Verdict
Berkeley	Physical things are ideas sustained in God's mind; laws are God's steady way of producing them	Filled [productive, nomic; personal; sustaining]
Kastrup's analytic idealism	The physical world is the outward appearance of one universal mind	Filled [constitutive], *disputed*: a critic can argue the universal mind relocates the question (§5.6)
Goff's cosmopsychism	The cosmos is conscious, and consciousness is the intrinsic nature of physical structure	Empty (identity), by clause (c)

View	Claim	Verdict
Many finite minds, no God	The physical is constructed from the experiences of many minds	Split (joint) if the minds are outside the order; Empty if they are part of it

14.8 The mathematical universe

Case. Max Tegmark (2008) holds that the physical universe *is* a mathematical structure.

Resolution. That is an identity, and clause (c) applies.

Verdict. Empty (identity). A different view, on which mathematical structures exist independently and *make* one of themselves physically real, would be Filled [nomic or constitutive; abstract]. The Pythagorean saying that "all is number" can be read either way.

14.9 Process theism

Case. For Whitehead (1929), God does not precede the world. God holds before every moment of the world its best possibilities, and so supplies order and direction. But "creativity," the fact that there are always new moments at all, is not God's product. Later process theologians say God creates "out of chaos," not out of nothing (Cobb and Griffin 1976).

Verdict. Split (law-side) [nomic, axiological; personal; sustaining]. Process theism is a living theistic tradition in Split, which shows the verdict is not only for ancient systems.

14.10 Gods inside the cosmos

Hesiod's gods are born from Chaos and Earth. Epicurus's gods are made of atoms. They are members of P*, and their worlds are **Empty**.

On a common reading of Latter-day Saint teaching, the elements are eternal (Doctrine and Covenants 93:33) and God organizes them. If that organizing is a non-internal ground* of order, the verdict is **Split (law-side)**; if God is held to be within the same lawful order as matter, it is **Empty**. *Disputed, and scored with caution:* this is an outsider's reading of a living tradition's texts.

14.11 Two principles

Manichaeism teaches two co-eternal principles, Light and Darkness, whose mixture is the physical world. **Split (joint).**

Some **Gnostic** systems have a lower craftsman who shapes matter, and above him a remote Monad from which everything, the craftsman included, ultimately comes. By the terminal clause the Monad is the occupant: **Filled**. If matter is independent of the Monad: **Split (two grounds)**.

14.12 Laws that evolve

C. S. Peirce (1891) suggested the laws of nature grew, as habits do. Lee Smolin (1997) proposed that universes reproduce through black holes, with slightly varied laws. Both explain lawfulness by a process inside P*, governed by some higher-order tendency. That tendency is internal. **Empty (regress or brute).**

14.13 What the hard cases show

Lesson	Cases
Bigger is not outside	Simulation, multiverse
Fundamental is not outside	Priority monism, the laws themselves
Divine is not the same as outside	Stoics, Hesiod, Epicurus
Outside need not be divine	Axiarchism
Outside need not be elsewhere	Spinoza, Advaita
The same word can land on either side of the line	“Mind” in the four idealisms; “God or Nature” in the Stoics and Spinoza

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Chapter 15 — Four case studies

Chapter 14 scored many views briefly. This chapter scores four in depth, one for each verdict where possible, following the protocol of §11.4. Quotations from Zhu Xi and Nāgārjuna are close paraphrases of standard English translations and should be checked against them.

15.1 Thomas Aquinas: Filled

The claims. Aquinas (→ A.8) holds the following in the *Summa Theologiae*.

- Everything other than God has its being from God (I, q. 44, a. 1).
- Creation is the coming forth of the whole of a thing’s being from the universal cause, with nothing presupposed (I, q. 45, a. 1). Only God creates (I, q. 45, a. 5).

- Creatures need God to keep them in being at every moment, as air needs the sun to stay lit (I, q. 104, a. 1).
- All things are directed by the *eternal law*, the plan of divine wisdom (I–II, q. 93).
- That the world had a beginning is held by faith and cannot be demonstrated (I, q. 46, a. 2).

Assent test. Aquinas affirms that something grounds the world. Passed.

Outside test. What does Aquinas say God is, in God's own nature? Not a body (I, q. 3, a. 1). Utterly simple, with no parts (q. 3, a. 7). A being whose essence is to exist (q. 3, a. 4). So God is not a member of P*. God is not a regular domain of states subject to laws, so the question is not relocated. And God is not identical to the world. Non-internal on all three clauses.

Unique test. One God (I, q. 11), who grounds* both existence (creation and conservation) and lawfulness (the eternal law). Because only God creates, there are no intermediate channels for the existence side.

Verdict. Filled [productive, nomic; personal; sustaining].

What the case teaches. Aquinas holds this verdict *while denying that reason can show the world began*. For him, dependence and beginning are separate matters. This is the historical source of the constraint that the ground is not an earlier event in time, and it is why Chapter 5 could set cosmology's question about a first moment to one side.

15.2 Nāgārjuna: Refused

The claims. In the *Mūlamadhyamakakārikā* (→ A.9), Nāgārjuna argues:

- Nothing arises from itself, from another, from both, or without a cause (1.1).
- A thing with intrinsic nature, *svabhāva*, could not be produced or depend on anything; but everything is dependent, so nothing has intrinsic nature (chapter 15).
- What arises dependently is what is called empty. That is a dependent designation, and it is the middle way (24.18).
- There are two truths, conventional and ultimate, and the ultimate is not taught without relying on the conventional (24.8–10).

In another work, the *Vigrahavyāvartanī*, he says that he has no thesis of his own to defend (verse 29).

Assent test. Offer Nāgārjuna the sentence "something outside the physical order grounds it." He rejects it: an ultimate ground would have to have intrinsic nature. Offer him "nothing grounds it; it is simply there." He rejects that too: arising without cause is one of the four options denied in the first verse, and he repeatedly distinguishes emptiness from nihilism. He affirms neither sentence. **Refused.**

Which presupposition? Totality (π_1): there is no self-standing whole. And determinacy (π_3): the four-cornered negation rejects "is," "is not," "both" and "neither."

A rival reading. Some interpreters read Madhyamaka as a positive thesis: reality is dependence all the way down, with no bottom level. If that is what Nāgārjuna asserts, he is saying that nothing outside grounds the order and that its internal dependence never terminates. That would be **Empty (regress)**. The dispute between this "metaphysical" reading and the "no-thesis" reading is a live one among scholars (compare Garfield 1995; Siderits and Katsura 2013; Westerhoff 2009).

Verdict. Refused (π_1, π_3) on the no-thesis reading; **Empty (regress)** on the metaphysical reading.

What the case teaches. The assent test turns a vague impression ("Buddhism doesn't do creation") into a precise question for interpreters: does Nāgārjuna *assert* groundlessness, or decline the question? It also shows why the Empty verdict must not be confused with Buddhist "emptiness."

15.3 Zhu Xi: three readings

The claims. In recorded conversations and letters (→ A.9; translated in Chan 1963, chapter 34), Zhu Xi says things that pull in different directions.

1. There has never been *qi* without *li*, or *li* without *qi*.
2. Asked which comes first, he says that fundamentally neither is before or after; but if one must trace their origin, one is obliged to say *li* is prior.
3. Before heaven and earth existed, there was, after all, only *li*.
4. *Li* has no will, no plan and no creative activity. It is *qi* that condenses and produces things.
5. The Supreme Ultimate is simply the *li* of heaven, earth and the myriad things.

Assent test. Zhu Xi affirms that *li* is that by which things are what they are. Passed.

Outside test. Is *li* non-internal? Statements 2 and 3 say *li* is prior and would be there without the physical world. On that reading it is like Plato's Forms, and non-internal. Statement 1 says *li* never exists apart from *qi*. On that reading it is like a universal that

exists only in its instances, which clause (c) of §5.5 counts as an aspect of the order. Internal.

Unique test, if *li* is non-internal. Does *li* ground* the existence of *qi*? Statement 4 suggests not: *li* does not produce. Other passages say that where there is this *li*, this *qi* comes to be.

Verdict. Three readings, three verdicts.

Reading	Verdict
Li is prior, and *qi* arises from it	Filled [nomic, productive; impersonal]
Li is prior and patterns *qi*, but does not produce it	Split (law-side) [nomic; impersonal]
Li is only the pattern found in *qi*	Empty (identity)

What the case teaches. The three readings are not an artefact of the framework. They correspond to real later positions. Thinkers such as Luo Qinshun and Wang Fuzhi moved toward the third reading, making *qi* basic. In Korea, the long debate between the schools of Yi Hwang and Yi I turned largely on the priority and activity of *li*. The framework locates a centuries-old East Asian controversy on the same map as the dispute between Plato and Aristotle about whether forms exist apart from things.

15.4 Graham Oppy: Empty

The claims. In *The Best Argument against God* (2013), Oppy compares two worldviews (→ A.16).

- **Naturalism:** natural reality is all of causal reality.
- **Theism:** natural reality was caused by God.

He argues that explanation must end somewhere on either view, and considers each kind of ending: an infinite regress, a necessary first item, or a contingent, unexplained first item. Whatever the theist says of God, the naturalist can say of the first natural state. Since naturalism then explains as much with fewer commitments, it is the better theory. Oppy's own preference is for a *necessary* initial state.

Assent test. Oppy affirms that nothing beyond natural reality accounts for it. Passed, on the "nothing" side.

Outside test. He cites nothing non-internal.

Verdict. Empty (necessary), with Empty (brute) and Empty (regress) as his stated alternatives.

What the case teaches. Oppy's argument is Chapter 4 put to work. He and Aquinas agree that explanation stops, and largely agree on the *kind* of stopping point: something necessary. They disagree on its *location*: inside the physical order, or outside it. With the two set side by side, the dispute is seen to be about simplicity and about whether a physical state is the sort of thing that can be necessary. It is not about cosmology, and it is not about whether one side "has an explanation" while the other lacks one.

Chapter 16 — A worked payoff: one debate, re-read

A framework earns its keep by making a real dispute clearer. This chapter applies God* to a well-known public debate.

16.1 The debate

In February 2014, at the Greer–Heard Forum in New Orleans, the philosopher William Lane Craig and the physicist Sean Carroll debated "God and Cosmology" (published in Stewart 2016). The summary below is a paraphrase of their main lines and should be checked against the published text.

- **Craig** argued that the universe began to exist, citing the Borde–Guth–Vilenkin theorem and thermodynamics; that whatever begins has a cause; and that the fine-tuning of the constants points to design.
- **Carroll** argued that the theorem does not establish a beginning and that there are consistent models with no first moment; that "cause" is not part of the vocabulary of fundamental physics, which deals in laws and patterns, so the universe can simply *be*; that a multiverse can account for fine-tuning; and that theism is too ill-defined to make predictions.

16.2 Scoring the positions

	Craig	Carroll
Assent test	Affirms an outside ground	Affirms that nothing outside is needed
Outside test	A timeless, immaterial, personal creator: non-internal	Cites only physical models: all internal
Unique test	One creator of matter and laws	—
Verdict	Filled [productive, nomic; personal]	Empty (brute)

Carroll's remark that "cause" is the wrong vocabulary is a reason of the π_2 kind. But he does not stop at declining the question. He asserts that the universe can just be. By the assent test of §9.1, that is Empty, not Refused.

16.3 What the framework reveals

Topic of the debate	What it bears on, by the framework
Did the universe have a first moment?	A question inside physics (§5.3). It matters to *one argument* for Filled, the kalām argument. It does not decide the verdict: Aquinas is Filled without a beginning, and a universe with a first moment can be Empty (brute).
Is "cause" applicable to the universe?	The real disagreement: π_2, and whether explanation may stop inside the order (Chapter 4).
Fine-tuning and the multiverse	The *specific* part of G-law: why these laws (§2.5). A multiverse is internal. It can answer "why these constants here?" It does not address why there is a lawful multiverse.
Is theism well-defined?	A complaint about equivocation, which the thin role answers (§7.5).

Three observations follow.

1. **Much of the debate concerned a question that neither verdict depends on.** Both speakers spent a large share of their time on whether the past is finite. On the framework's analysis, a finite past is neither necessary nor sufficient for Craig's verdict, and an infinite past is neither necessary nor sufficient for Carroll's.
2. **The decisive disagreement was touched only briefly.** It is whether a brute stopping point inside the physical order is acceptable. That is a dispute about the principle of sufficient reason, not about cosmology.
3. **Each could have granted the other more.** Carroll could grant a beginning and remain Empty (brute). Craig could grant a beginningless multiverse and remain Filled, as Aquinas would. Seeing this would have moved the debate to its real centre.

Key idea. The framework does not say who won. It says what the debate was about, which is the more useful thing to know.

Chapter 17 — Do scorers agree? A study design

A scoring scheme is useful only if different people, applying it independently, reach the same result. This chapter designs a test of that. **The study has not been run.** It is presented as the next piece of work, and as the feature that would most distinguish this project from its neighbours, none of which reports such a test.

17.1 Design

Element	Plan
Materials	12 passages of 150–300 words, each a primary text or close paraphrase, with author names removed where possible
Passages	Three expected in each verdict, drawn from Chapter 18: for example Aristotle, Udāyana, Whitehead; Lucretius, Oppy, the Stoics; Nāgārjuna, Carnap, Kant; Aquinas, Plotinus, Leslie
Scorers	At least six: three with training in philosophy, three without
Instructions	The one-page core (§11.1) and the protocol (§11.4); no access to the atlas
Task	For each passage: a verdict, and optionally the extension tags

Element	Plan
Measure	Fleiss' kappa, a standard statistic for agreement among several raters beyond chance, computed separately for verdicts and for tags

17.2 Predictions, stated in advance

1. Agreement on the four **verdicts** will be substantial: kappa of at least 0.6, on the conventional scale of Landis and Koch (1977).
2. Agreement on **tags** will be lower than on verdicts.
3. Disagreements will cluster on the **Empty–Refused** boundary and on **constitutive** grounds (Spinoza-like cases), the two places Chapters 5 and 9 identified as hardest.

17.3 What would count against the framework

- Kappa below 0.4 on verdicts would show that the three tests are not clear enough to use.
- Disagreements scattered evenly, not clustered, would show that the difficulty is general and not confined to known boundary cases.
- Trained and untrained scorers diverging sharply would show the framework is not, in practice, a *shared* vocabulary.

Stating the failure conditions in advance is what makes this a test and not a demonstration.

Chapter 18 — An atlas of views

The table scores about fifty views with the protocol. "/" separates rival readings. *Disputed* marks an entry where specialists disagree about the reading itself. Every entry is open to correction by better textual evidence.

View	Verdict	Basis	See
Atomism (Democritus, Epicurus, Lucretius)	Empty (brute)	Eternal atoms and void; gods inside the cosmos	§12.1
Heraclitus	Empty (identity)	The logos is the world's own measure	§12.1

View	Verdict	Basis	See
Pythagoreans	Empty (identity) / Filled [abstract]	“All is number”	§14.8
Plato, *Timaeus* reading	Split (two grounds)	Craftsman and Forms; receptacle ungrounded	§12.2
Plato, *Republic* reading	Filled [axiological, nomic]	The Good as source of being	§12.2
Aristotle	Split (law-side)	Unmoved mover; eternal world	§12.3
Stoicism	Empty (identity)	A bodily, immanent God	§14.2
Hesiod; Homeric religion	Empty	Gods born within the cosmos	§14.10
Plotinus	Filled [productive; impersonal]	Emanation from the One	§12.4
Genesis 1	Filled / Split (law-side)	Creation from nothing, or ordering of chaos	§12.5
Rabbinic Judaism; Maimonides	Filled [productive, nomic; personal]	Creation	§12.5
Classical Christianity (Augustine, Aquinas)	Filled [productive, nomic; personal; sustaining]	Creation, conservation, eternal law	§15.1
Ash’arī occasionalism (al-Ghazālī)	Filled [productive, nomic; personal; sustaining]	Regularities are God’s custom	§2.3
Avicenna	Filled [productive, by necessary emanation]	The Necessary Existent	§12.5
Sikhism	Filled [productive, nomic; sustaining]	*Ik Onkar*	§12.5
Nāsadīya Sūkta	A profile, not a verdict	“Perhaps he does not know”	§9.5
Advaita Vedānta (Śaṅkara)	Filled [constitutive, productive], *disputed*	*Brahma Sūtra* 1.1.2; levels rule	§14.5

View	Verdict	Basis	See
Rāmānuja	Filled [constitutive; personal]	The world as Brahman's body	§12.6
Madhva	Filled [productive; personal]	A distinct creator	§12.6
Classical Sāṃkhya	Empty (brute)	Eternal primal matter	§12.6
Nyāya–Vaiśeṣika	Split (law-side) [personal]	A Lord over eternal atoms	§12.6
Mīmāṃsā	Empty	Beginningless world; no creator	§12.6
Jainism	Empty	Uncreated, eternal universe	§12.6
Early Buddhism	Refused (π_2)	The undetermined questions	§12.6
Dharmakīrti	Empty	Argues against a creator	§12.6
Madhyamaka (Nāgārjuna)	Refused (π_1, π_3) / Empty (regress), *disputed*	No thesis, or dependence without end	§15.2
Daodejing, cosmogonic reading	Filled [productive, constitutive; impersonal]	Chapter 42	§12.7
Daodejing, apophatic reading	Refused (π_3)	Chapter 1	§12.7
Guo Xiang	Empty (brute)	Things are self-so	§12.7
Zhu Xi	Filled / Split (law-side) / Empty (identity), *disputed*	Priority of *li*	§15.3
Manichaeism	Split (joint)	Two eternal principles	§14.11
Gnostic systems	Filled / Split (two grounds)	The Monad, and the status of matter	§14.11
Descartes; Boyle; Newton	Filled [productive, nomic; personal]	Laws as divine decree	§13.1

View	Verdict	Basis	See
Spinoza	Filled [constitutive, nomic; impersonal] / Empty (identity)	Standard and naturalist readings	§14.2
Leibniz	Filled [productive, nomic, axiological; personal]	The best possible world	§13.1
Classical deism	Filled [productive, nomic; personal; once]	A creator who does not intervene	§10.5
Hume	Empty / a sceptical profile, *disputed*	The *Dialogues*	§13.2
Kant	Refused (knowledge); Filled (practical postulate)	Two claims, two scores	§9.5
Hegel	Filled [constitutive] / Empty (identity), *disputed*	Right and left readings	§13.2
Whitehead; process theism	Split (law-side) [nomic, axiological; personal]	Creativity is not God's product	§14.9
Tillich	Filled [constitutive]	Being-itself	§9.5
Logical positivism (Carnap, Ayer)	Refused (π_2, π_3)	Not a factual question	§9.2
Wittgenstein, *Tractatus*	Refused (π_2), *disputed*	"The riddle does not exist" (6.5); but see 6.44	A.21
Russell (1948)	Empty (brute)	"Just there"	§9.4
Latter-day Saint teaching, a common reading	Split (law-side) / Empty, *disputed*	God organizes eternal elements	§14.10
Analytic theism (Swinburne, Plantinga, Craig, Pruss)	Filled [productive, nomic; personal]	Creator and sustainer	§16.2
Oppy	Empty (necessary)	A necessary initial state	§15.4
Carroll	Empty (brute)	The universe simply is	§16.2

View	Verdict	Basis	See
Schaffer's priority monism	Empty (identity)	The whole is fundamental	§14.3
Leslie; Rescher	Filled [axiological; abstract]	Ethical requiredness	§14.6
Berkeley	Filled [productive, nomic; personal]	Ideas in God's mind	§14.7
Kastrup	Filled [constitutive], *disputed*	Mind-at-large	§14.7
Goff's cosmopsychism	Empty (identity)	Mind as the inner nature of the physical	§14.7
Tegmark	Empty (identity)	The universe is a mathematical structure	§14.8
Peirce; Smolin	Empty (regress or brute)	Laws from an internal process	§14.12
Bostrom's simulation hypothesis	Verdict-neutral	Enlarges P*	§14.1

Figure 18.1 · The atlas on the verdict map; views with rival readings appear un…

What the atlas shows.

1. **All four verdicts are well populated.** Split is not a curiosity. It holds Aristotle, Nyāya, one reading of Plato, and a living school of theism.
2. **The verdicts cut across "religious" and "secular."** A Stoic and a physicist share Empty. A Thomist and an axiarchist with no god share Filled.
3. **Disjunctive and disputed entries mark old controversies.** Plato, Zhu Xi, Spinoza, Hegel, Nāgārjuna and Śaṅkara are exactly the figures whose interpreters have argued longest. The framework does not settle those arguments. It states them in a common form.

Part V — Position and limits

Chapter 19 — The scholarly neighbours, and what is new

None of the ingredients of God* is new. This chapter names the nearest existing work, family by family, and then says exactly what the framework adds. Entries here were recalled from memory and must be checked before citation.

19.1 "God" as a title or role

- **Nelson Pike (1969)** read "God" as a title. His title carried the full perfect-being job description. God* keeps the title form and replaces the job description with a thin, cosmological one.
- **David Lewis (1970)** supplied the general method for role terms (§7.2).

19.2 Ground and foundation

- **Paul Tillich** called God the "ground of being." God* can express his view, and does not require it.
- **Kenneth Pearce (2017)** argues that the argument from contingency is best understood with God as the *ground*, in the metaphysician's sense, of contingent reality, not its cause. **Einar Bohn (2018)** develops a related "divine foundationalism." These are the closest work to Chapter 3. They defend one species of dependence as the right account of God and world. God* is neutral among species, because its purpose is comparison, not advocacy.
- **Recent debate books** frame theism against naturalism as a dispute about the *foundation* of reality: Oppy (2013; 2018), Rasmussen and Leon (2019), Oppy and Pearce (2022). God* sharpens their shared question and extends it beyond two contestants.
- **Richard Swinburne (2004)** treats the argument from the existence of a physical universe separately from the argument from its orderliness. That is the clearest precedent for two sides. For Swinburne they are two pieces of evidence for one hypothesis. God* makes them two sides of a role, and so can say that they may have different grounds.

19.3 Comparing ultimates across traditions

- **The Comparative Religious Ideas Project** (Neville 2001) used deliberately *vague categories*, such as "ultimate realities," which each tradition specifies in its own way. God* is a cousin, but more exact: it returns one of four verdicts, and a verdict can be

wrong.

- **Wesley Wildman (2010; 2017)** sorts "ultimacy models" by the *nature* of the ultimate: an agent, a ground of being, a process. God* sorts by whether and how a *role* is filled. The two classifications cross-cut and can be used together.
- **Diller and Kasher (2013)** edited a very large survey, *Models of God and Alternative Ultimate Realities*, with chapters on classical theism, pantheism, panentheism, process thought, ground-of-being views and non-theistic ultimates. **Buckareff and Nagasawa (2016)** edited *Alternative Concepts of God*. These collections are the natural source material for the atlas. They present models side by side. They do not supply a single question with a closed set of answers against which every model is scored. That is what God* attempts.
- **John Hick (1989)** proposed that the great traditions are responses to one "Real." That is a substantive thesis, religious pluralism. God* is only a question, and is neutral on pluralism.
- **J. L. Schellenberg (2005)** defined *ultimism*: there is a reality ultimate in being, in value, and in the good it offers. God* isolates a relative of his first dimension, restricted to the physical order and divided into two sides.

19.4 Global philosophy of religion

Kevin Schilbrack (2014), the essays in Draper and Schellenberg (2017), and Yujin Nagasawa's projects have urged philosophers to stop treating Christian theism as the default case. God* is offered as a working tool for that programme.

Figure 19.1 · Neighbouring projects and where God* sits (placed by quadrant, no…

19.5 What is new

Contribution	How new
1. A two-sided role with a named **Split** verdict	Partly new. The two explananda are old. Making them sides of one role, with a verdict for their separation, is not standard.
2. The **assent test** separating Refused from Empty	New in this literature, as far as the author knows

Contribution	How new
3. The **relocation test** for what counts as outside the physical order	Partly new. It turns Albert's point against Krauss into a general rule and extends it to simulations and constitutive grounds.
4. **Ground***: neutrality among species of dependence, with a required tag	Partly new. It applies Wilson's pluralism to a comparative purpose.
5. Scoring **claims**, not traditions or thinkers	A methodological rule in the spirit of comparative religion
6. A stated **core**, separate from extensions, and a **reliability test** with failure conditions	New for projects of this kind

Key idea. God* is best described as an instrument, not a theory. It is a precise shared question, a closed set of answers, and tests for applying them. Whether it is a *good* instrument is an empirical matter that Chapter 17 says how to check.

19.6 Terms that clash with established usage

1. **"Internal."** Carnap (1950) used "internal" and "external" for questions asked inside or about a linguistic framework. Here "internal" means inside the physical order. In formal writing, "intra-physical" avoids the confusion.
2. **"Ground."** In current metaphysics "grounding" is a specific non-causal relation. This book writes **ground*** for its broader relation.
3. **"Empty."** The Empty verdict is not Buddhist emptiness (*śūnyatā*), which belongs under Refused.

Chapter 20 — Objections and replies

20.1 "This is the cosmological argument in disguise."

Objection. The role is shaped like the conclusion of an argument for God. The game is rigged.

Reply. The role is shaped like the *question* that argument answers. Naturalists ask the same question and answer "nothing." The framework assumes no principle of sufficient reason, counts Empty (brute) as a complete answer, and gives refusal a verdict of its own. The opposite complaint is also heard: that the thin role strips theism of everything theists care about. A definition accused of bias from both directions is probably near the middle.

20.2 “The distinctions are too fine to apply.”

Objection. With enough tags, any verdict can be defended for any view.

Reply. That is why the core is separated from the extensions (Chapter 11), and why Chapter 17 proposes a test with stated failure conditions. If independent scorers do not agree on the four verdicts, the objection is correct.

20.3 “What grounds the ground?”

Objection. If an outside ground needs no ground, why does the physical order need one?

Reply. That is exactly the dispute between Filled and Empty, and the framework leaves it open. Chapter 4 showed that every answer places a stopping point somewhere. Whether the occupant is necessary, self-explaining or brute is part of the occupant theory.

20.4 “Religions are not sets of propositions.”

Objection. Philosophers in the tradition of Wittgenstein, such as D. Z. Phillips (1976), hold that religious language often expresses a way of life and makes no metaphysical claims. Scoring it as theory distorts it.

Reply. The framework scores claims. Where there is no claim about the ground of the physical order, there is no score (§8.4). That is different from Refused. And where a tradition contains both a practice and a body of doctrine, only the doctrine is scored, and the score says nothing about the practice.

20.5 “Who are you to say what a tradition teaches?”

Objection. The atlas reads traditions from outside.

Reply. The protocol requires a cited text for every entry, a named reading, a disjunctive verdict where readings differ, and a statement of what evidence would overturn the entry. Entries on living traditions are marked for caution. The atlas is meant to be corrected by people who know the texts better.

20.6 “Why only the physical?”

Objection. Minds and numbers may be non-physical. Why not ask for the ground of everything?

Reply. Because nothing is outside everything, so the widest role cannot be filled by definition (§11.3). The physical order is the one domain all parties grant. Wider roles are available as extensions.

20.7 “The whole framework assumes that such questions have answers.”

Objection. Deflationists about metaphysics, such as Amie Thomasson (2015), hold that questions like this are either easy or empty. The framework is realist from the start.

Reply. The framework states its presuppositions (§6.4) and lets a view reject them. A deflationist is scored Refused, on π_2 or π_3, and is not misdescribed. What the framework cannot do is be *written* from a deflationist standpoint. That is a real limit, noted in Chapter 21.

20.8 “The name will put people off.”

Objection. “God,” even starred, alienates atheists and non-theists, and may annoy believers.

Reply. The star marks a stipulated term. The word was chosen because the dispute it names is the one people ordinarily carry on with the word “God.” Where the name distracts, **G*** or “the ground role” does the same work.

Chapter 21 — What God* cannot do

1. **It does not say which verdict is true.** It supplies a question and its possible answers. The arguments for each answer stand or fall on their own.
2. **It does not capture what makes a religion a religion.** Worship, ethics, community and salvation lie outside the role by design. A tradition’s verdict is one small fact about it.
3. **The physical/non-physical line may be culturally specific.** Some Indigenous and animist ontologies do not divide the world that way. For such a view the right outcome may be “not scored.” Forcing a verdict would repeat the mistake the framework exists to prevent.
4. **It is written from inside a realist picture of explanation.** It can score those who reject that picture. It cannot share their starting point.
5. **It is all-or-nothing about grounding.** A view on which something outside *partly* or *probably* accounts for the order has no graded verdict.
6. **The boundary between realizer and constitutive ground remains hard** (§5.6). The relocation test locates the dispute and does not always end it.
7. **Its reliability is untested.** Until the study of Chapter 17 is run, the claim that scorers will agree is a prediction.

Chapter 22 — Conclusion

22.1 What was built

Chapter	Piece added
1	There is a question at the top of the ladder of “why”
2	It has two sides: G-existence and G-law
3	“Because” means ground*, in one of four species
4	Every answer places a stopping point, inside or outside
5	“Outside” means non-internal: not a member of P*, not a relocation, not identical
6	“The whole” is the physical things taken together; three presuppositions are stated
7	“God*” is the title of the role; the role is thin
8–11	Three tests yield exactly four verdicts; a small core, with optional extensions

Figure 22.1 · The definition, assembled

22.2 The final statement

> **God*** is the title of a role, not the name of a being. The role is filled by whatever, if anything, is the **terminal non-internal ground*** of the **physical order as a whole**: of the fact that there is such an order (**G-existence**), and of the fact that it is lawful (**G-law**).
>
> **Ground*** is asymmetric explanation. It may be productive, constitutive, nomic or axiological. It need not come earlier in time, and it need not make its result inevitable.
>
> The **physical order** is everything linked to ordinary physical things by space and time, by lawful interaction, or by lawful realization.
>
> Something is **non-internal** if it is not part of that order, is not the order or an aspect of it, and is not a further regular domain that only moves the question.
>
> A view gives one of four verdicts: SERF.
>
> - **Split:** outside grounding, but no single ground of both sides.
> - **Empty:** nothing outside grounds either side.
> - **Refused:** the view will affirm neither that something grounds the order nor that nothing does.
> - **Filled:** one such ground of both sides.
>
> Whether the occupant is personal, good, necessary, or acts once or always belongs to the **occupant theory**, not the role. Not knowing is a profile, not a verdict. What makes no claim is not scored.

22.3 Closing

The question “why is there a lawful world at all?” has been asked in Greek, Sanskrit, Pali, Chinese, Hebrew, Arabic and Latin, and answered in many ways. The word “God” has too often stood between those answers and kept them from being compared. God* does not answer the question. It holds the question still, so that a Thomist, a Madhyamaka Buddhist, a Stoic, a Spinozist and a physicist can each say exactly what they think sits in the keystone slot — or why nothing does, or why there is no slot — and be understood by the others. That is a modest aim, and it is achievable.

Appendix A — Primers

Each primer gives the background the main text assumes. Dates for ancient figures are approximate.

A.1 The Presocratics (sixth to fifth centuries BCE)

The first Greek philosophers lived around the Aegean before Socrates. Their works survive only as fragments quoted by later writers.

- **Thales of Miletus** (c. 585 BCE) said everything is, or comes from, water. What mattered was the idea that one natural principle underlies everything.
- **Anaximander** replaced water with the *apeiron*, the boundless or indefinite.
- **Heraclitus of Ephesus** (c. 500 BCE) taught that everything flows, like a river you cannot step into twice, but that the flux obeys a *logos*, a rational proportion.
- **Parmenides of Elea** (c. 475 BCE) argued that what-is cannot come from what-is-not.
- **The atomists** Leucippus and Democritus (fifth century BCE) held that reality is indivisible bodies moving in void. **Epicurus** (341–270 BCE) and the Roman poet **Lucretius** (*On the Nature of Things*, c. 55 BCE) made atomism an ethical philosophy that frees people from fear of the gods, who exist but take no part in the world.

A.2 Agrippa and ancient skepticism

Agrippa, probably of the first century CE, is known through the later skeptic Sextus Empiricus. He listed “modes” for suspending judgment. Three of them — infinite regress, circularity and arbitrary assumption — form the trilemma of §4.1. Hans Albert later named it the “Münchhausen trilemma,” after the baron who claimed to have pulled himself from a swamp by his own hair.

A.3 Plato: the craftsman, the Forms and the Good

Plato (c. 428–348 BCE) was Socrates’ student and Aristotle’s teacher. He wrote dialogues and founded the Academy in Athens.

The Forms. Plato held that beyond the changing things we see there are eternal, unchanging Forms: Beauty itself, Justice itself, Equality itself. Particular things are what they are by “participating” in the Forms. A beautiful face is beautiful by sharing in Beauty. The Forms are known by reason, not by the senses. In the *Republic*’s allegory of the cave, ordinary life is watching shadows on a wall, and philosophy is turning round toward the Forms and at last the sun.

The Good. In the *Republic* (Book VI, 508–509), the Form of the Good is compared to the sun. The sun makes things visible and makes them grow. The Good makes the Forms knowable and gives them their being, while itself being "beyond being in dignity and power."

The *Timaeus* and the Demiurge. In the *Timaeus* (c. 360 BCE), a speaker tells a "likely story" of how the cosmos came to be. A good divine craftsman, the *dēmiourgos* ("artisan"), wanted everything to be as good as possible. He looked to an eternal model, the Forms, and brought order to a pre-existing, chaotic "receptacle." He shaped it with geometry: the elements are built from triangles. The craftsman is not all-powerful. He persuades *necessity*; he does not overrule it.

Why it matters here. Plato offers three candidates: the craftsman (a maker and orderer), the Forms (a pattern) and the Good (a value). None of them creates the receptacle. That is why Plato is Split or Filled depending on how the three are related. Readers since antiquity have debated whether the *Timaeus* describes a creation in time or is a teaching device for eternal dependence.

A.4 Aristotle: four causes, an eternal world, an unmoved mover

Aristotle (384–322 BCE) studied at Plato's Academy for twenty years, tutored Alexander the Great, and founded the Lyceum.

- **"Metaphysics."** The treatises now called the *Metaphysics* were Aristotle's "first philosophy": the study of being as being, and of the highest causes.
- **The four causes** (*Physics* II): material, formal, efficient and final.
- **The eternal world.** Aristotle argued that time and motion have no beginning.
- **The unmoved mover** (*Metaphysics* Λ). Everything that moves is moved by something. To avoid an endless regress there must be a mover that is not moved: pure actuality, "thought thinking itself." It moves the heavens "as the object of love moves," as a final cause.

A.5 The Stoics and Plotinus

The Stoics, founded by Zeno of Citium around 300 BCE and including Chrysippus, Seneca, Epictetus and Marcus Aurelius, were materialists: everything real is a body. They identified God with the active principle in matter, a fiery, intelligent breath (*pneuma*), the logos. The cosmos periodically dissolves into fire and is reborn in the same order.

Plotinus (204/5–270 CE), founder of Neoplatonism, taught in Rome. His *Enneads* describe reality as flowing from **the One**, a principle so simple that no description fits it. From the One "overflows" **Intellect** (containing the Forms), then **Soul**, then the physical world. The emanation is necessary and unchosen, like light from the sun, and the One is not diminished by it. Neoplatonism shaped Augustine, Pseudo-Dionysius, Islamic philosophy and medieval thought.

A.6 Augustine

Augustine of Hippo (354–430 CE) was a North African bishop, a Manichaean and then a Neoplatonist before his conversion. In *Confessions* Book XI he asks what God was doing before creating the world, and answers that there was no "before": God created time with the world. He also taught creation from nothing and continuous conservation.

A.7 Avicenna, al-Ghazālī and Maimonides

- **Ibn Sīnā (Avicenna)** (980–1037), a Persian polymath, wrote *The Healing*. He distinguished essence (what a thing is) from existence (that it is). A triangle's definition does not tell you whether any triangle exists. Anything whose essence does not include existence needs a cause of its existing. To avoid a regress there must be a **Necessary Existent**, whose essence is its existence. The world flows necessarily from it through a hierarchy of intellects.
- **Abū Ḥāmid al-Ghazālī** (1058–1111), in *The Incoherence of the Philosophers*, attacked Avicenna's necessary emanation and eternal world. In the Seventeenth Discussion he argued that fire does not *necessarily* burn cotton. God creates the burning on the occasion of contact, according to God's custom. This is *occasionalism*.
- **Moses Maimonides** (1138–1204), Jewish philosopher and jurist, wrote *The Guide of the Perplexed*. He argued that neither creation nor an eternal world can be demonstrated, but that creation is the more plausible and is what the Torah teaches.

A.8 Thomas Aquinas

Thomas Aquinas (1225–1274), a Dominican friar, joined Aristotle's philosophy to Christian doctrine.

- **The Five Ways** (*Summa Theologiae* I, q. 2, a. 3) argue to God from motion, efficient causation, contingency, degrees of perfection, and the goal-directedness of natural things.
- **God is subsistent being itself.** God's essence is to exist.
- **Creation is not a change.** It is the total dependence of a thing's being on God.

- In ***On the Eternity of the World*** (c. 1270) he argued that a created world that always existed is not a contradiction. That the world began is known by faith.

A.9 Indian, Chinese and other traditions

- **The Vedas** (c. 1500–500 BCE) are the oldest Sanskrit scriptures. **Ṛta** is the cosmic and moral order. The **Nāsadīya Sūkta** (*Rig Veda* 10.129), the "Hymn of Creation," begins "Then was neither non-being nor being" and ends in open uncertainty.
- **The Upaniṣads** (c. 800–200 BCE) teach that **Brahman**, ultimate reality, is identical with **ātman**, the deepest self.
- **Sāṃkhya** (classical form in Īśvarakṛṣṇa's *Sāṃkhyakārikā*, c. 350–450 CE) is dualist and non-theistic. Countless conscious **puruṣas** witness the unfolding of **prakṛti**, primal matter.
- **Nyāya-Vaiśeṣika** is the school of logic and atomism. **Udāyana** (c. 975–1050) argued that composite things need an intelligent maker, so the world needs **Īśvara**, who arranges eternal atoms.
- **Mīmāṃsā**, the school of Vedic interpretation (Kumārila, seventh century), argued against a creator. The Veda is authorless and the world beginningless.
- **Jainism** (Mahāvīra, sixth to fifth centuries BCE) holds that the universe is uncreated and eternal.
- **Buddhism** (the Buddha, fifth century BCE) teaches *dependent origination*: everything arises from conditions. The Buddha left some questions "undetermined" (*avyākṛta*), including whether the world is eternal. **Dharmakīrti** (seventh century) argued against a creator.
- **Nāgārjuna** (c. 150–250 CE) founded **Madhyamaka**, the "Middle Way." His *Mūlamadhyamakakārikā* argues that everything is empty (*śūnya*) of intrinsic nature (*svabhāva*). The *catuṣkoṭi*, or four-cornered analysis, considers "is," "is not," "both" and "neither," and often rejects all four.
- **Vedānta** comprises the schools that interpret the Upaniṣads, the *Bhagavad Gītā* and the *Brahma Sūtras*. **Śaṅkara** (c. 700–750), *Advaita*, "non-dual": only Brahman is ultimately real. **Rāmānuja** (traditionally 1017–1137), *Viśiṣṭādvaita*: souls and matter are the real body of a personal God. **Madhva** (1238–1317), *Dvaita*: God, souls and matter are eternally distinct.
- **Sikhism** was founded by Guru Nānak (1469–1539). The *Mūl Mantar* opens the Guru Granth Sahib: one creator, true, without fear or enmity, timeless, unborn, self-existent.

- **Daoism.** The ***Daodejing*** is attributed to Laozi (compiled c. fourth century BCE). The ***Zhuangzi*** is a book of parables and arguments. **Guo Xiang** (d. 312 CE), in his commentary on it, argued that things are "self-so" (*ziran*) and nothing produces them.
- **Neo-Confucianism. Zhu Xi** (1130–1200) shaped the orthodoxy of later imperial China. Everything is **li** (principle, pattern) and **qi** (vital stuff, material force). The **taiji**, the Supreme Ultimate, is the totality of li. Later critics, including **Luo Qinshun** (1465–1547) and **Wang Fuzhi** (1619–1692), gave priority to qi. In Korea, **Yi Hwang** (T'oegye) and **Yi I** (Yulgok) led rival schools on the relation of li and qi.
- **Manichaeism** was founded by Mani (third century CE). Light and Darkness are two eternal principles, and the world is their mixture.
- **Gnosticism** names a family of movements of the early Christian era in which a lower craftsman made the material world and the true God is a remote Monad.

A.10 Spinoza and Leibniz

Baruch Spinoza (1632–1677), a Dutch-Jewish philosopher and lens grinder, was expelled from his synagogue in 1656. His *Ethics* (1677) proceeds "in geometric order," with definitions, axioms and propositions. There is one substance, God or Nature, with infinitely many attributes. He distinguishes *natura naturans*, nature as active source, from *natura naturata*, nature as what follows from it.

Gottfried Wilhelm Leibniz (1646–1716), a German polymath, co-invented the calculus. He is known for the **principle of sufficient reason**, for the view that God chose the **best of all possible worlds**, and for *monads*. In *On the Ultimate Origination of Things* (1697) he argued that even an endless series of states, each explained by the one before, needs a reason outside the series. His example is an eternal succession of geometry books, each copied from the last.

A.11 Descartes, Boyle, Newton, Leibniz and Clarke, and deism

- **René Descartes** (1596–1650) derived the laws of motion from God's immutability (*Principles of Philosophy*, 1644, Part II).
- **Robert Boyle** (1627–1691) described laws as God's rules for matter.
- **Isaac Newton** (1643–1727), in the *General Scholium* (1713), argued that the system of sun, planets and comets could only come from an intelligent and powerful being.
- **The Leibniz–Clarke correspondence** (1715–16) set Leibniz against Samuel Clarke, who spoke for Newton.

- **Deism**, a movement of the seventeenth and eighteenth centuries that included John Toland, Matthew Tindal and Thomas Paine, accepted a creator known by reason and rejected revelation and miracles.

A.12 David Hume

David Hume (1711–1776), the Scottish empiricist, argued that we never observe necessary connections between causes and effects, only constant conjunction. His *Dialogues concerning Natural Religion* (published 1779) stage a debate between Cleanthes, who defends the design argument; Demea, who defends an a priori argument from contingency; and the sceptic Philo. In Part IX it is Cleanthes who answers Demea with the twenty-particles argument. “Commit it then to the flames” is the last line of Hume’s *Enquiry concerning Human Understanding* (1748).

A.13 Kant and Hegel

Immanuel Kant (1724–1804), in the *Critique of Pure Reason* (1781; 2nd ed. 1787), argued that the mind structures experience through space, time and categories such as causality, and that these apply only within possible experience. When reason tries to think the world as a completed whole it produces **antinomies**: equally good proofs that the world has a beginning and that it has none. In the *Critique of Practical Reason* (1788) he held that morality requires us to *postulate* God, freedom and immortality.

G. W. F. Hegel (1770–1831) described reality as the self-development of **Spirit**, which externalizes itself as nature and returns to itself in history, art, religion and philosophy. “Right” Hegelians read this theistically. “Left” Hegelians such as Feuerbach and Marx read it naturalistically.

A.14 Tillich and Heidegger

Paul Tillich (1886–1965), a German-American Protestant theologian, taught in *Systematic Theology* (vol. 1, 1951) that God is not a being alongside others but **being-itself**, the **ground of being**. To argue that God “exists,” he held, already makes God a thing among things.

Martin Heidegger (1889–1976) opened *Introduction to Metaphysics* (lectures of 1935) with “Why are there beings at all instead of nothing?”

A.15 Whitehead and process philosophy

Alfred North Whitehead (1861–1947), co-author with Russell of *Principia Mathematica*, developed a “philosophy of organism” in *Process and Reality* (1929). Reality consists of momentary “actual occasions” of experience. **Creativity** is the ultimate principle by which

many become one new occasion. God has a *primordial nature*, which holds all possibilities and draws the world toward order and value, and a *consequent nature*, which receives the world. Charles Hartshorne, John Cobb and David Ray Griffin developed **process theism**.

A.16 Contemporary philosophy of religion

- **Alvin Plantinga** (b. 1932) revived the ontological argument in modal form and defended the rationality of belief in God.
- **Richard Swinburne** (b. 1934) argued in *The Existence of God* that theism is probable on the total evidence.
- **William Lane Craig** (b. 1949) revived the *kalām* argument: whatever begins to exist has a cause; the universe began to exist; so it has a cause.
- **Alexander Pruss** and **Timothy O'Connor** defend versions of the argument from contingency.
- **Graham Oppy** (b. 1960), an Australian naturalist, compares theism and naturalism as whole worldviews.
- **Kenneth Pearce** defends a version of theism on which God grounds contingent reality.

A.17 Grounding theory

- **Kit Fine** ("Guide to Ground," 2012) treats grounding as an operator on sentences.
- **Jonathan Schaffer** ("On What Grounds What," 2009) treats metaphysics as the study of what grounds what, and defends *priority monism*.
- **Gideon Rosen** (2010) connects grounding with reduction and essence.
- **Jessica Wilson** (2014) argues that "Grounding" is too coarse and should give way to specific relations.

A.18 Cosmologists

- **Georges Lemaître** (1894–1966), a Belgian priest and physicist, proposed the "primeval atom," later called the Big Bang.
- **Edwin Hubble** (1889–1953) measured the expansion.
- **Stephen Hawking** (1942–2018) and **Roger Penrose** (b. 1931) proved the singularity theorems. Hawking and **James Hartle** (1939–2023) proposed the no-boundary state.
- **Alan Guth** (b. 1947) proposed inflation.
- **Alexander Vilenkin** (b. 1949) proposed tunnelling from nothing and co-proved the Borde–Guth–Vilenkin theorem.

- **Sean Carroll** (b. 1966) is a cosmologist who defends naturalism in public debate and in print.
- **Max Tegmark** (b. 1967) proposed the mathematical universe hypothesis.

A.19 The Russell–Copleston debate

In 1948 the BBC broadcast a debate on the existence of God between the philosopher **Bertrand Russell** and the Jesuit historian of philosophy **Frederick Copleston**. Copleston defended Leibniz's argument from contingency. Russell replied that the universe is "just there, and that's all," and that while every man has a mother, the human race has not. The debate is reprinted in Russell's *Why I Am Not a Christian* (1957).

A.20 Philosophers of language

- **Gottlob Frege** (1848–1925) distinguished a term's *sense* from its *reference*.
- **Bertrand Russell** (1872–1970), in "On Denoting" (1905), analysed definite descriptions.
- **Frank Ramsey** (1903–1930), a Cambridge prodigy who died at 26, showed how to state a theory's content without its theoretical terms.
- **P. F. Strawson** (1919–2006), in "On Referring" (1950), argued that failed presuppositions leave statements without a truth value.
- **Saul Kripke** (1940–2022), in *Naming and Necessity* (1980), argued that names are rigid designators and that some necessary truths are known only by experience.

A.21 Logical positivism, Carnap and Wittgenstein

The Vienna Circle of the 1920s and 1930s, including **Rudolf Carnap** (1891–1970), and A. J. Ayer's *Language, Truth and Logic* (1936) held that a statement is meaningful only if it is true by definition or checkable by experience. Carnap's "Empiricism, Semantics, and Ontology" (1950) distinguished *internal* questions, asked within a framework, from *external* questions about the framework.

Ludwig Wittgenstein (1889–1951) ends the *Tractatus* (1921) with "whereof one cannot speak, thereof one must be silent." Proposition 6.44 says the mystical is not *how* the world is but *that* it is. Proposition 6.5 says "the riddle does not exist." Whether the book dismisses the question or reveres it is disputed.

A.22 The revival of metaphysics

- **W. V. O. Quine** (1908–2000), in "On What There Is" (1948), made ontology respectable again.

- **Kripke** restored talk of necessity and essence.
- **David Lewis** (1941–2001) built a systematic metaphysics, including the best-system theory of laws and the method for defining theoretical terms.

A.23 The "why anything?" literature

- **Robert Nozick** (*Philosophical Explanations*, 1981, ch. 2) surveyed possible answers.
- **Derek Parfit**'s "Why Anything? Why This?" (1998) separated the two questions and took axiarchism seriously.
- **John Leslie** (*Value and Existence*, 1979) and **Nicholas Rescher** (*Axiogenesis*, 2010) defend axiarchism. **Tim Mulgan** (2015) and **Philip Goff** (2023) explore related views on which the universe has a purpose without a traditional God.
- **Peter van Inwagen** (1996) argued that there being something is overwhelmingly probable, since there are infinitely many ways for there to be something and one way for there to be nothing.
- **Bede Rundle** (2004), **Jim Holt** (2012) and **Sean Carroll** (2018) are further treatments; Holt's is written for general readers.

A.24 Idealism, simulation and the mathematical universe

- **George Berkeley** (1685–1753), an Irish bishop, held that to be is to be perceived.
- **Bernardo Kastrup** (2019) defends analytic idealism.
- **Philip Goff** (2017) defends cosmopsychism.
- **Nick Bostrom** (2003) gave the simulation argument.
- **David Chalmers** (2022) argues that simulated worlds are real.

A.25 Comparative and global philosophy of religion

- **John Hick** (1922–2012): the great traditions respond to one "Real."
- **Robert Cummings Neville** (b. 1939) directed the Comparative Religious Ideas Project with Wesley Wildman.
- **Wesley Wildman** (b. 1961) develops comparative religious philosophy and "ultimacy models."
- **J. L. Schellenberg** (b. 1959) proposes ultimism.
- **Jeanine Diller and Asa Kasher**, and **Andrei Buckareff and Yujin Nagasawa**, edited the two main collections on models of God and alternatives to them.
- **Kevin Schilbrack** argues that philosophy of religion should study all religions and their practices.

Appendix B — The original infographic, annotated

The framework was first set out as a one-page infographic (Figure B.1). Seven of the figures in Chapters 5, 7, 9 and 10 are panels taken from it.

God* — a shared definition

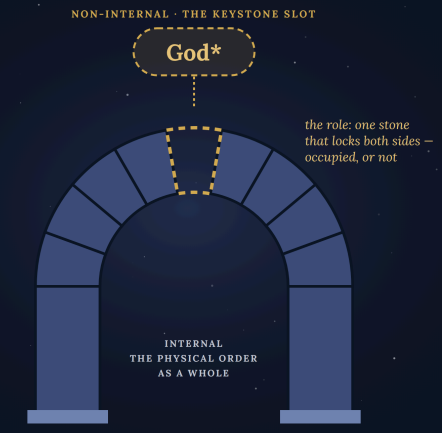
Built so rival metaphysical theories can be tested against the same job.

THE DEFINITION

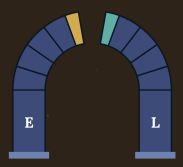
God* is the title of a role, not the name of a being. The role is occupied by whatever, if anything[1], is the non-internal[2] ground[3] of the physical order as a whole[4].

That whole has two sides. **G-existence:** that there is a physically structured domain at all, rather than no such domain. **G-law:** that this domain is lawful — regular and structured, not a heap of disconnected happenings.

God names one non-internal ground of both. If only one side has such a ground, or the two grounds are different things, God* is not uniquely filled. Say: existence-ground here, law-ground there.*



Four answers below: the slot filled, split, empty, or refused.



S SPLIT

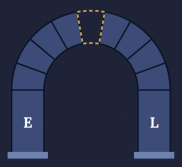
two grounds

Existence and lawfulness do not share one outside account.

The domain may be brute and the laws grounded — or the reverse; or two different grounds; or a starter after which law needs no further ground.

MAPS[5]
Some deist pictures · some Platonic pictures (craftsman and Forms as two jobs)

NOTE
If the two grounds differ, do not call both God*. The title is for one ground of both sides.



E EMPTY

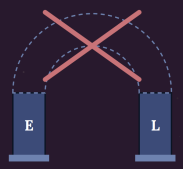
no ground

The physical order has no outside account. That includes 'it has always been' and 'it just is.'

Empty accepts the question and answers: nothing fills it. A beginningless or brute physical whole still sits here. Ignorance is not this verdict.

MAPS[5]
Atheism / philosophical naturalism · classical Samkhya · Buddhist schools that deny a creator without treating the question as confused

NOTE
Internal processes that are fundamental among other internal processes do not occupy the role. A later physics theory does not, by itself, fill the role.



R REFUSED

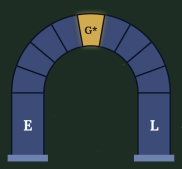
no role

The role itself is the wrong question. There is not a missing occupant; the job description does not apply.

Refused declines the question. Empty says nothing fills the role. Refused says the role misfires. Score the claim, not the tone.

MAPS[5]
Some Madhyamaka readings · some Taoist readings that refuse the Tao as a ground of a whole

NOTE
The definition remains usable as vocabulary even when a tradition rejects the job. Refused is not a fifth kind of occupant.



F FILLED

one ground

One thing outside the physical order is why the order is there and why it is lawful.

One non-internal ground covers both G-existence and G-law. A highest being inside the cosmos does not count. Personhood is an occupant theory, not the role.

MAPS[5]
Judaism · Christianity · Islam · Sikhism · theistic Hinduism (Ishvara; some Vedanta when Brahman grounds the world)

NOTE
Occupant theories are not the role: person, will, goodness, revelation, worship, Trinity, covenant, Quran, avatar. Filled can be personal or not.

INTERNAL VS NON-INTERNAL
Something is internal if stating it already assumes a physical order: spacetime, laws, fields, particles, vacuum, history, or a generator that takes an ordered domain as input. Big Bang, inflation, bounce, quantum gravity, dark energy, and a multiverse generator are internal unless they can be stated without that assumption. An earlier stage of the same order is still internal.

CONSTRAINTS
Ground is not an earlier event in time, not a first particle, and not a piece of the order. Being fundamental among physical processes does not fill the role. Necessity given the laws does not fill the role. Personhood and worship-worthiness are claims about an occupant, not part of the role. Not knowing is unfinished inquiry; it is not Empty and not a fifth verdict.

NOTES [1] If anything — Empty is a legal answer. [2] Internal — already presupposes spacetime, laws, or a physical system.



Figure B.1 · The original one-page infographic

Element	What works	What this book changes, and where
Title and subtitle	"Built so rival metaphysical theories can be tested against the same job" states the purpose exactly	No change
Definition sentence	Title, not name; "if anything" allows a vacant role	"ground" becomes ground* (§3.4); "terminal" is added (§10.2)
G-existence and G-law	Two sides with long separate histories (Part III)	Each side gains a general and a specific part (§2.5); the number two is defended (§2.6)
Gold note on Split	Anticipates the verdict and names it well	No change
Keystone arch	A structural ground, not an earlier event	Optional corbelled arch for Empty (§7.6)
Split panel	A named verdict that neighbouring frameworks lack	Deism moves to Filled (§10.5); Aristotle, Nyāya and process theism are added
Empty panel	"Ignorance is not this verdict"; fundamental is not a ground	Four kinds of Empty (§9.3); the Stoics added
Refused panel	"Score the claim, not the tone"	The assent test (§9.1) and three reasons to refuse (§9.2)
Filled panel	Occupant theories kept out; "personal or not"	Impersonal and non-theistic examples added
Internal vs non-internal box	The right idea, with good examples	Three clauses: member, relocation, identity (§5.5); the simulation case
Constraints box	Ground is not an earlier event; not knowing is not a verdict	"Once or always is not part of the role"; "what makes no claim is not scored"

Element	What works	What this book changes, and where
Footnote 5	Tags show where a tradition often lands, not every adherent	Extended into the rule of scoring claims (§8.4)

Overall. The infographic's central moves were right from the start: a title and not a name, a thin role, two sides, four verdicts, Refused kept apart from Empty, and occupant theories kept out of the role. Everything this book adds is precision.

Recommended changes for a second version.

1. **Definition.** Write "ground*" and footnote its four species. Add "terminal."
2. **Internal box.** Replace the test with the three clauses: member, relocation, identity. Add the simulation example.
3. **Split panel.** Change the map to "Aristotle · Nyāya · process theism · Plato's *Timaeus*." Move classical deism to Filled, and keep "a starter" under Split only for a starter of one side.
4. **Empty panel.** Add four small labels: brute, necessary, regress, identity. Add the Stoics as an example of Empty with a deity. Consider drawing a corbelled arch.
5. **Refused panel.** Add the assent test in one line: "Will affirm neither 'something grounds it' nor 'nothing does.'" Keep Madhyamaka here, with a footnote that the Empty verdict is not Buddhist emptiness.
6. **Filled panel.** Add "Plotinus · Spinoza · axiarchism" to show that Filled need not be personal or even theistic.
7. **Constraints box.** Add "Once or always is not part of the role" and "What makes no claim is not scored."
8. **Mnemonic.** Label the four panels SERF.

Appendix C — Glossary

Term	Meaning
Archē	Greek: origin, first principle
Assent test	Will the view affirm that something grounds the order, or that nothing does? If neither: Refused

Term	Meaning
Asymmetric	If A relates to B, B does not relate to A in the same way
Axiarchism	The view that the world exists because it is good that it should
Brute fact	A fact with no explanation
Channel	An intermediate link between the terminal ground and the physical order
Contingent	True, but could have been false
Corbelled arch	An arch of overhanging stones with no keystone
Creation from nothing	Creation with no pre-existing material
Deism	Belief in a creator known by reason, without revelation or intervention
Demiurge	Plato's divine craftsman
Emanation	The flowing of reality from a source, as in Plotinus
Equivocation	Using one word in two senses within one argument
Explanandum, explanans	What is explained; what explains it
Fine-tuning	The apparent sensitivity of life to the values of physical constants
G-existence	The fact that there is a physical order, with its inventory and initial state
G-law	The fact that the order is lawful, with the laws it has
God*	The title of the role "terminal non-internal ground* of the physical order as a whole"
Ground*	Asymmetric explanation: productive, constitutive, nomic or axiological
Grounding	In metaphysics, non-causal "in virtue of" dependence

Term	Meaning
Hempel’s dilemma	“Physical” defined by current physics is false; by ideal physics, empty
Internal	A member of P*, a relocation of the question, or identical to P* or an aspect of it
Intrinsic-description rule	Test what a view says an item is in itself, not a label that relates it to the world
Levels rule	Score a multi-level tradition at the level where it places the physical order
Li and qi	Neo-Confucian principle or pattern; material force
Logos	Greek: word, reason, rational order
Modal collapse	The worry that the principle of sufficient reason makes every truth necessary
Necessary	Could not have been otherwise
Occasionalism	God is the only true cause; creatures are occasions
Occupant theory	Claims about what fills the role: personal, good, and so on
P*	The maximal physical order: everything linked to ordinary physical things by space and time, lawful interaction, or lawful realization
Plural logic	Logic that speaks of many things together without positing a collection
Presupposition	What a question takes for granted
Principle of sufficient reason	Everything has an explanation
Profile	A person’s degrees of confidence across the four verdicts
Ramsey sentence	A theory restated with its special terms replaced by variables
Realization	One thing being made up of, or running on, the states of another

Term	Meaning
Relocation	Citing a further regular domain, which raises the same two-sided question again
Rigid designator	A term that picks out the same thing in every possible situation
Ṛta	Vedic cosmic order
Side	A fact about the physical order that is common ground, global and distinct
Svabhāva	Sanskrit: intrinsic, self-standing nature
Terminal ground	The outside ground not itself grounded* by a further outside item
Title	A term naming an office, not an individual

Appendix D — From this book to a paper or a doctorate

D.1 An honest assessment

This book is not a doctoral thesis. A thesis is original research written over several years under supervision and examined by a university. This book shows the shape such a project could take. Its core is **publishable as a journal article** if written tightly, argued against the literature in Chapter 19, and checked reference by reference.

D.2 Path 1: a journal article

A suggested structure, about 9,000 words:

1. The equivocation problem, with the Stoics and Tillich as opening examples (700 words).
2. The role: title terms, the Ramsey–Lewis method, thinness (1,000).
3. Two sides, and why two (1,000).
4. Ground*: why strict grounding and causation are each biased (1,400).
5. The relocation test and the simulation case (1,300).
6. Four verdicts and the assent test (1,400).

7. Comparison with Pike, Pearce, Swinburne, Neville and Wildman, Diller and Kasher, Schellenberg (1,400).
8. Objections and conclusion (800).

Journals to consider. *Religious Studies*, *Sophia*, *International Journal for Philosophy of Religion*, *European Journal for Philosophy of Religion*, *Faith and Philosophy*, and *Philosophy East and West* for the comparative angle. Check each journal's current scope and guidelines.

How submission works.

- Most philosophy journals review anonymously. Remove your name, and submit to one journal at a time.
- Decisions often take three to nine months.
- Rejection or "revise and resubmit" is the usual result, even for established authors. The referee reports are valuable.
- Posting a preprint on PhilArchive establishes priority and invites comment.
- Independent scholars do publish. Engagement with the literature matters more than affiliation.

D.3 Path 2: the reliability study as a second paper

The study designed in Chapter 17 could be run with volunteers and a spreadsheet. Its results, whatever they are, would make a short second paper, and would be unusual in this field.

D.4 Path 3: doctoral study

- **Requirements.** Usually a bachelor's degree and often a master's; a writing sample, for which a polished God* paper would serve; a statement of purpose; letters of recommendation.
- **Where.** Departments strong in philosophy of religion, metaphysics or comparative philosophy. Read recent faculty publications before applying, and look for people working on the authors in Chapter 19.
- **A bridge.** A master's degree, or individual courses, supply the background and the letters.

D.5 Path 4: public routes

- A general-audience essay of 2,500–4,000 words.

- A talk or poster at a philosophy of religion or religious studies meeting.
- An email to a researcher cited here, with a two-page summary, the infographic, and one specific question.
- The atlas as an open resource that specialists in each tradition are invited to correct.

D.6 A twelve-month plan

Months	Goal	Output
1–2	Read tiers 1 and 2 below; verify every reference in this book	Notes; a checked bibliography
3–4	Draft the article	First full draft
5	Feedback from two or three readers	Revised draft
6	Preprint; submit to a first-choice journal	Submission
6–9	Run the reliability study	Data and a short report
9–12	Revise the infographic; respond to referees	Version 2 infographic; resubmission

D.7 A reading ladder

- **Tier 1, start here.**
  - Holt, *Why Does the World Exist?*
  - The Russell–Copleston debate.
  - Plato, *Timaeus* 27d–34b.
  - Albert's review of Krauss.
- **Tier 2.**
  - Pike (1969) and Lewis (1970).
  - Parfit (1998).
  - Oppy, *The Best Argument against God*.
  - Swinburne, *The Existence of God*, the chapters on the cosmological and teleological arguments.
  - Schellenberg, *Prolegomena*, chapter 1.
- **Tier 3.**
  - Fine, "Guide to Ground"; Wilson, "No Work for a Theory of Grounding."

- Pearce (2017) and Bohn (2018).
- Pruss, *The Principle of Sufficient Reason*.
- Diller and Kasher; Buckareff and Nagasawa.
- Neville, *Ultimate Realities*; Wildman, *In Our Own Image*.
- Garfield's Nāgārjuna; Deutsch on Advaita; Chan's *Source Book*, the Zhu Xi chapter.

Appendix E — Changes from the first edition

Topic	First edition	This edition
Order of exposition	Background, then framework, then repairs ("rulings")	One construction; each piece built once
"Internal"	Membership, vocabulary, identity	Membership (with a realization link), **relocation**, identity; the intrinsic-description rule
Simulation case	Claimed solved, but the definition of P* did not cover it	Solved by the realization link and the relocation test
Refused versus Empty	Three presuppositions; an "R→E" fallback notation	The **assent test** decides; presuppositions classify the reasons; the fallback notation is dropped
Russell (1948)	Refused with fallback Empty	Empty (brute)
Number of sides	Asserted	Defended by three conditions for being a side
Split	Subtypes with subscripts, including an endless-chain case	Defined as outside grounding without a unique occupant; plain-word forms; boundary cases flagged
Notation	Letters and subscripts	Plain words in brackets
Core and extensions	Mixed together	Separated
Nāgārjuna	Refused	Refused or Empty (regress), by reading

Topic	First edition	This edition
Zhu Xi	Filled or Split	Split, Empty (identity) or Filled, by reading
Case studies	Two short scorecards	Four in depth
Payoff	None	A real debate re-read (Chapter 16)
Reliability	Not addressed	A study design with failure conditions (Chapter 17)
Literature	Missed several close neighbours	Adds Pearce, Bohn, Diller and Kasher, Buckareff and Nagasawa
Corrections	The twenty-particles argument attributed loosely; 2 Maccabees called an explicit statement of the doctrine	Attributed to Cleanthes; May's caution noted

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Author–date style. **Every entry was compiled from memory, not from a database.** Titles and years are believed correct; volume, issue and page details in particular must be checked against the publications before any formal use.

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